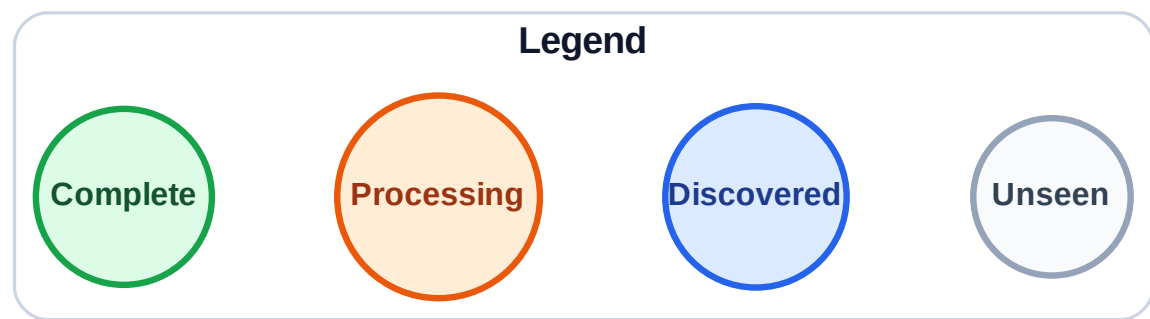
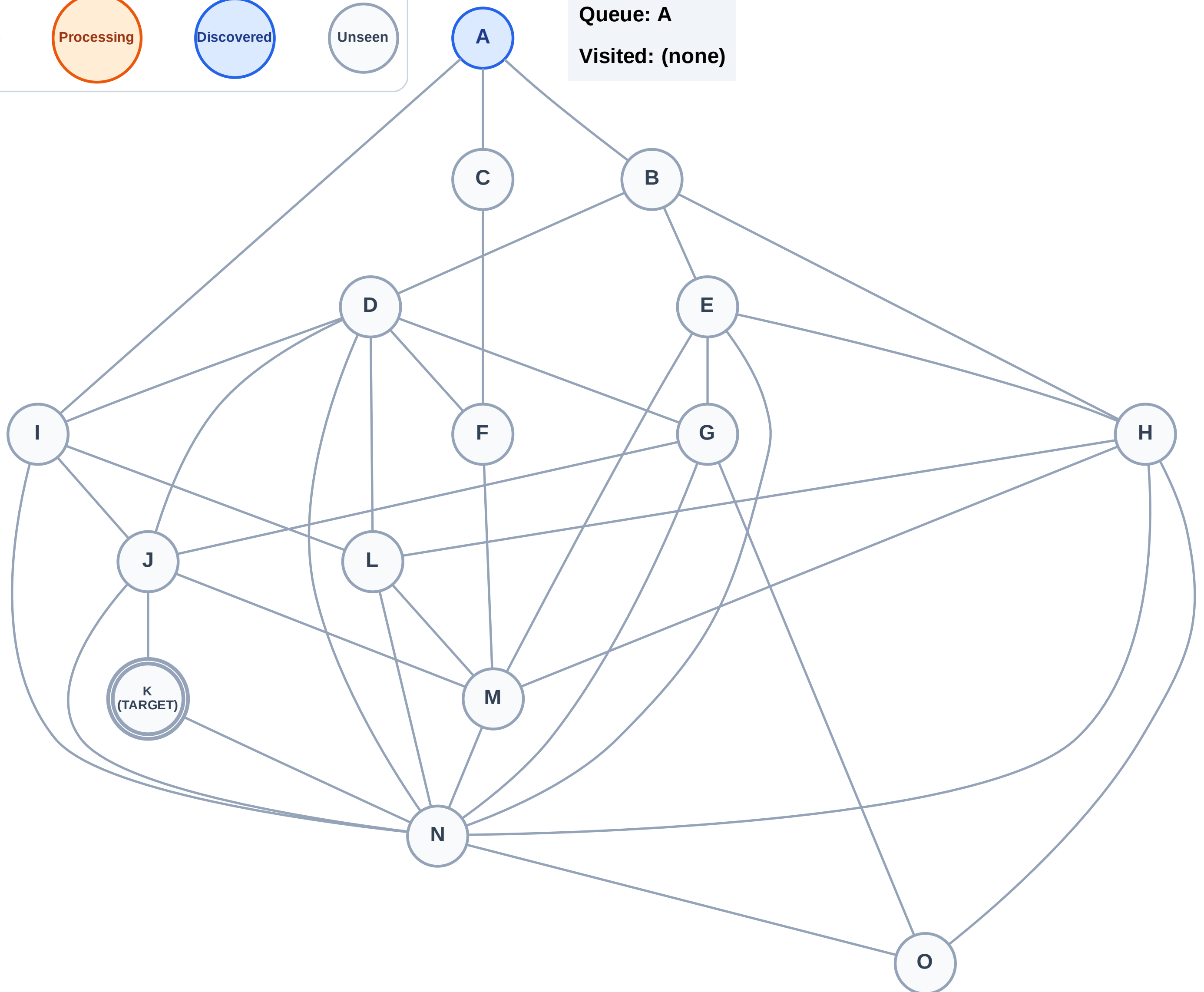


BFS Traversal

Step 1: Start at A (Seeking target node K)



Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

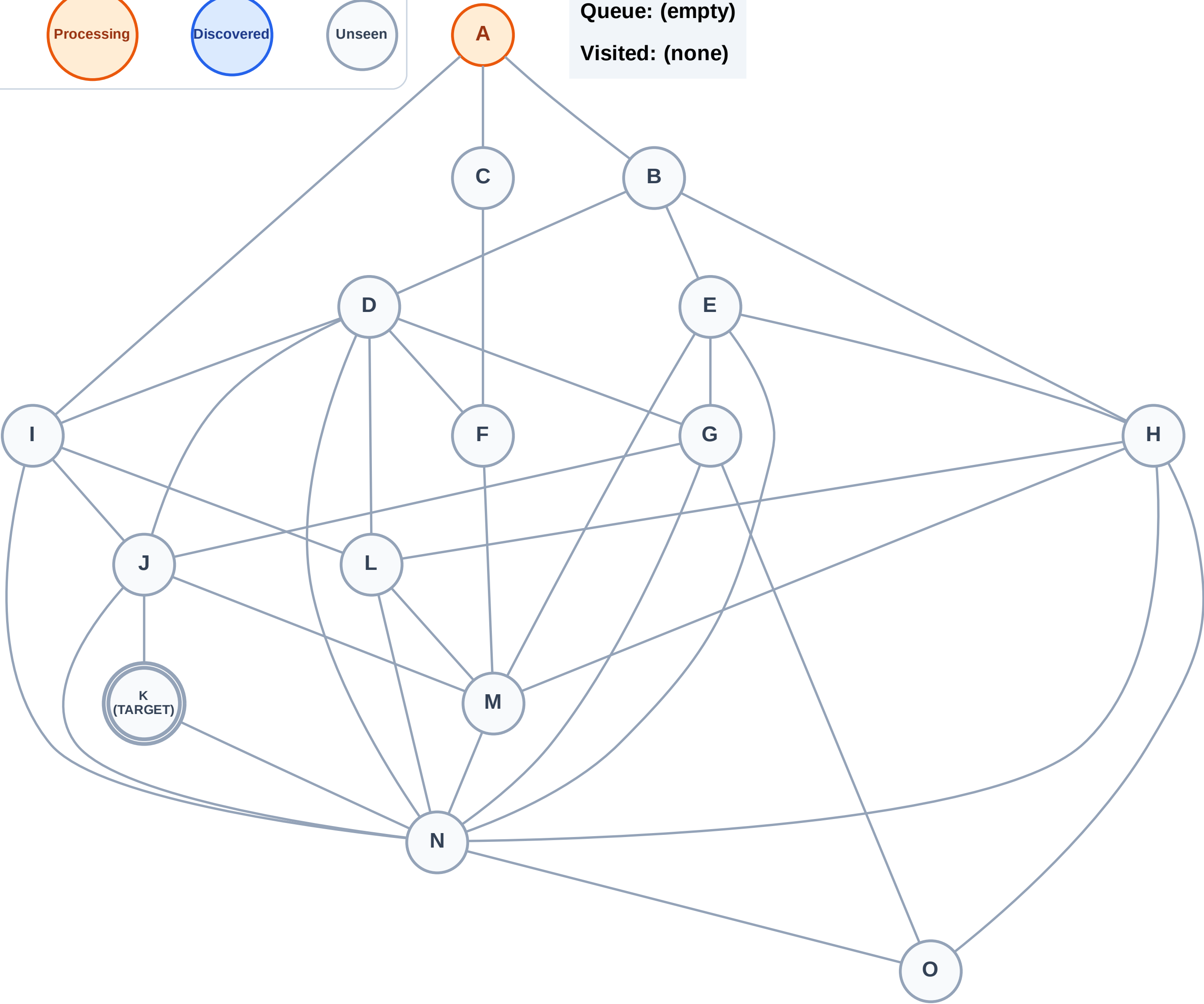
Complete

Processing

Discovered

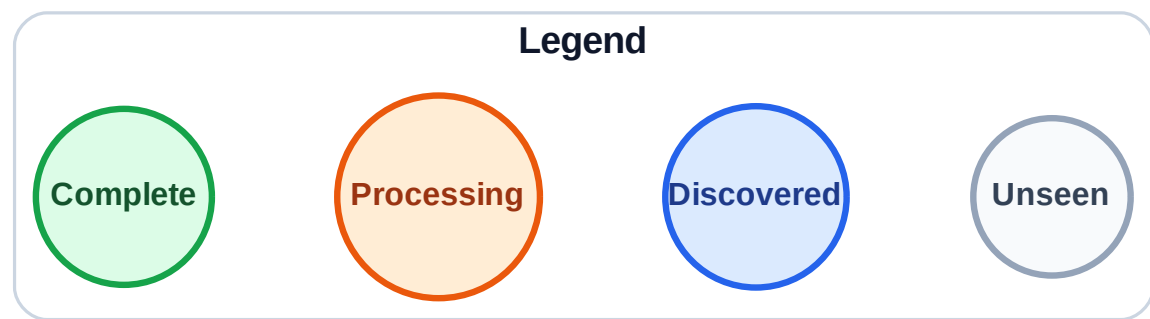
Unseen

Queue: (empty)
Visited: (none)



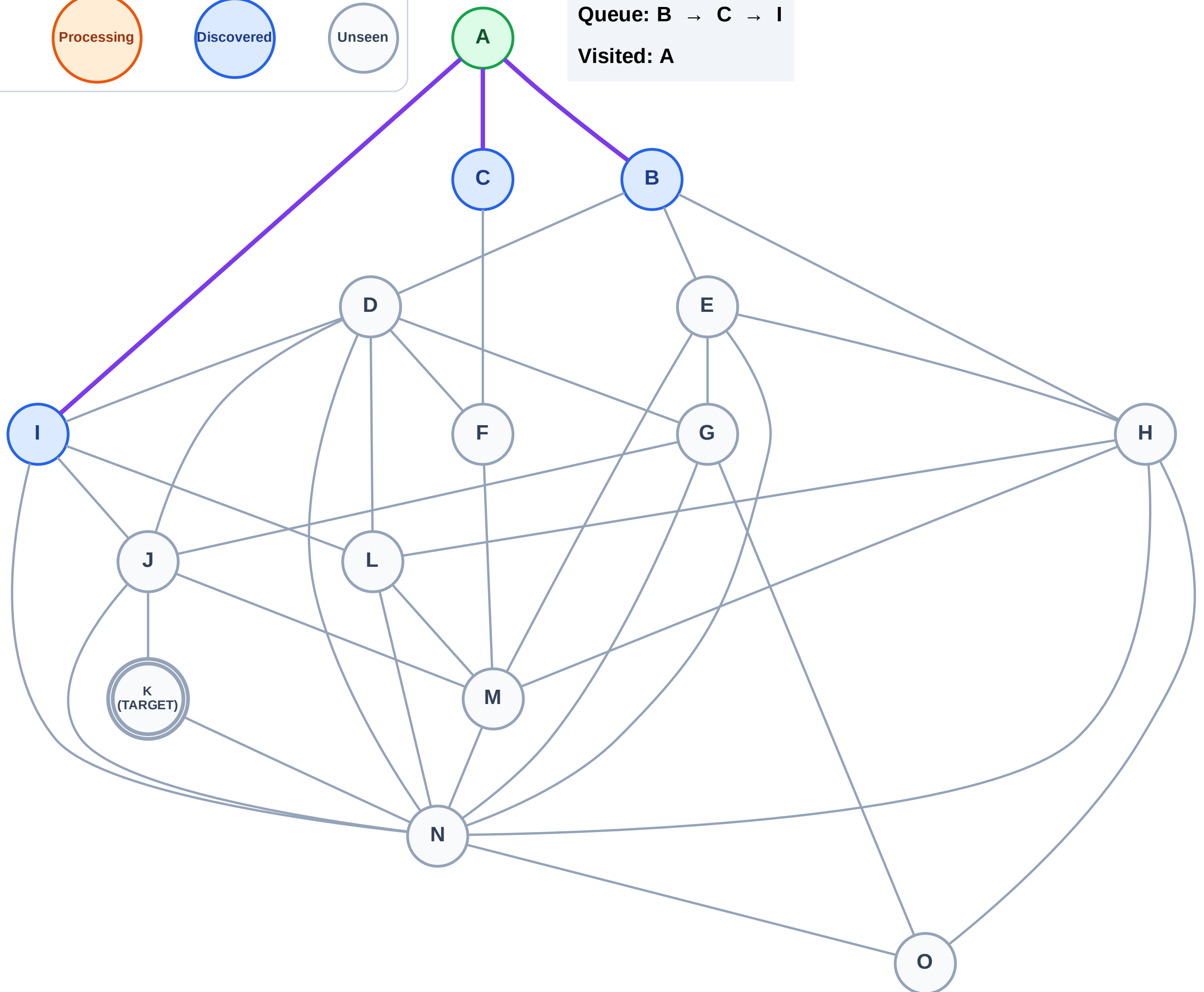
BFS Traversal

Step 3: Discover B, C, I from A



Queue: B → C → I

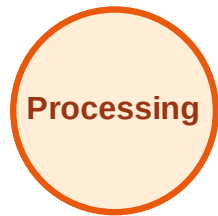
Visited: A



BFS Traversal

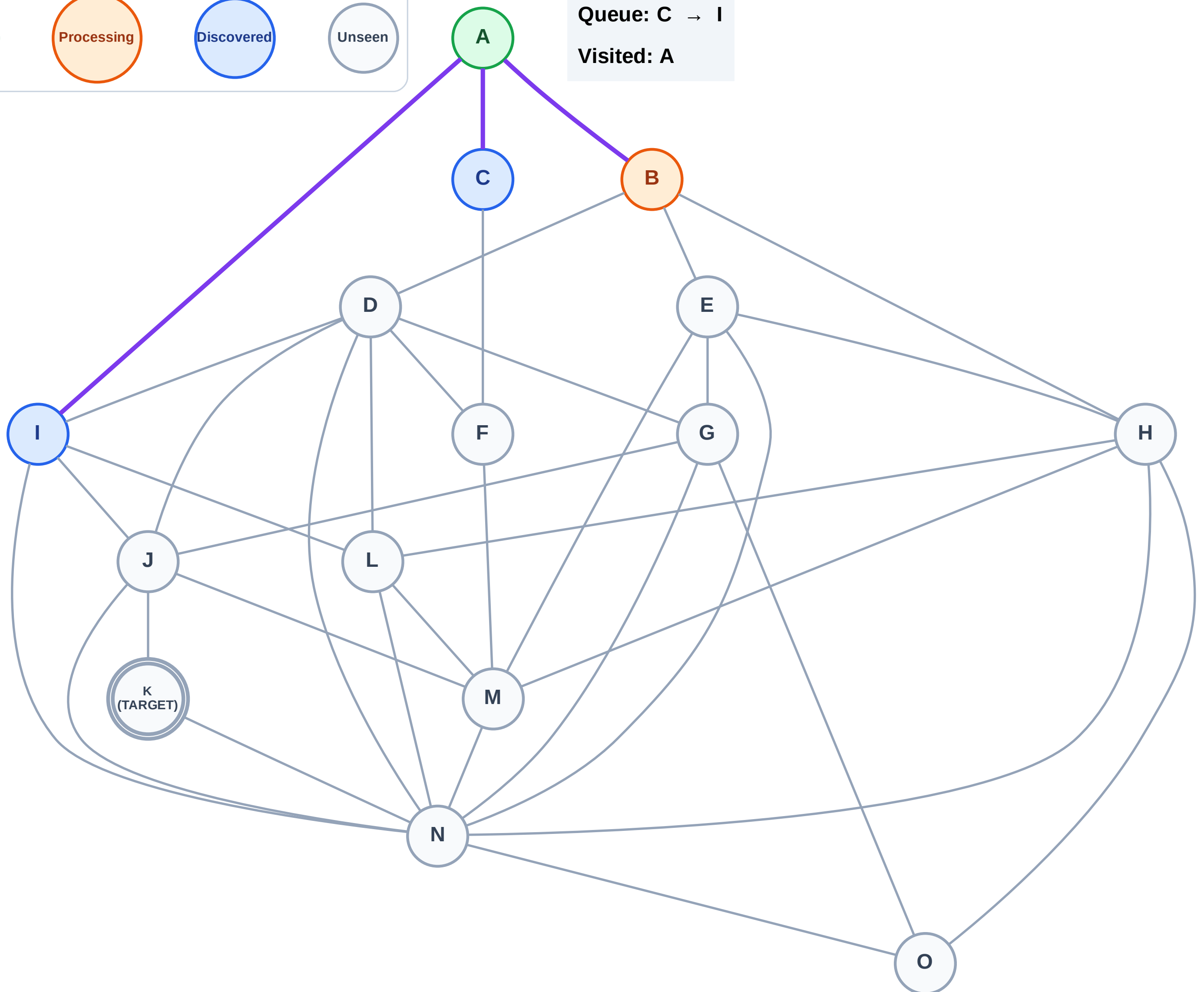
Step 4: Process node B

Legend



Queue: C → I

Visited: A

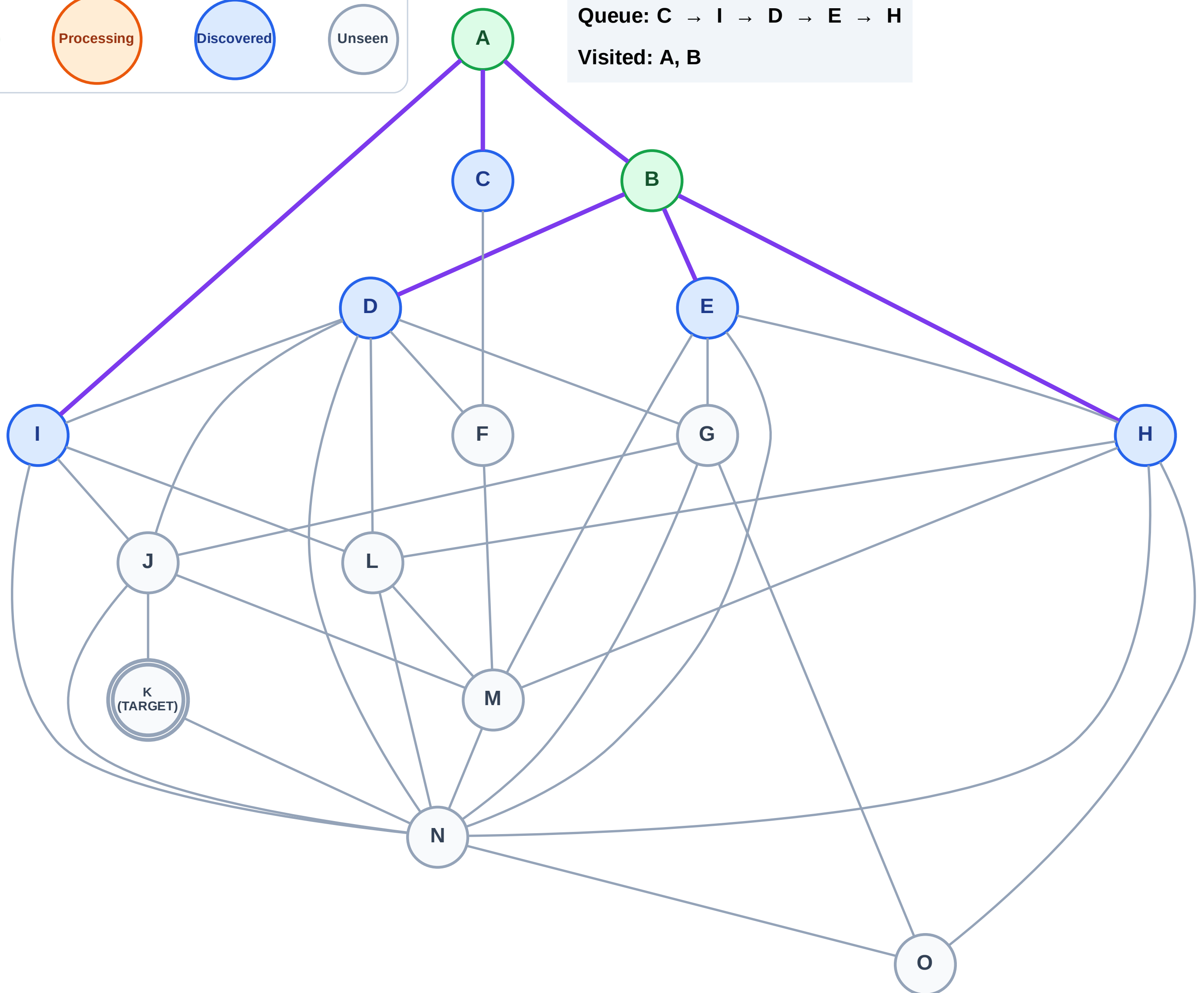


Step 5: Discover D, E, H from B

Queue: C → I → D → E → H

Legend

Unseen



BFS Traversal

Step 6: Process node C

Legend

Complete

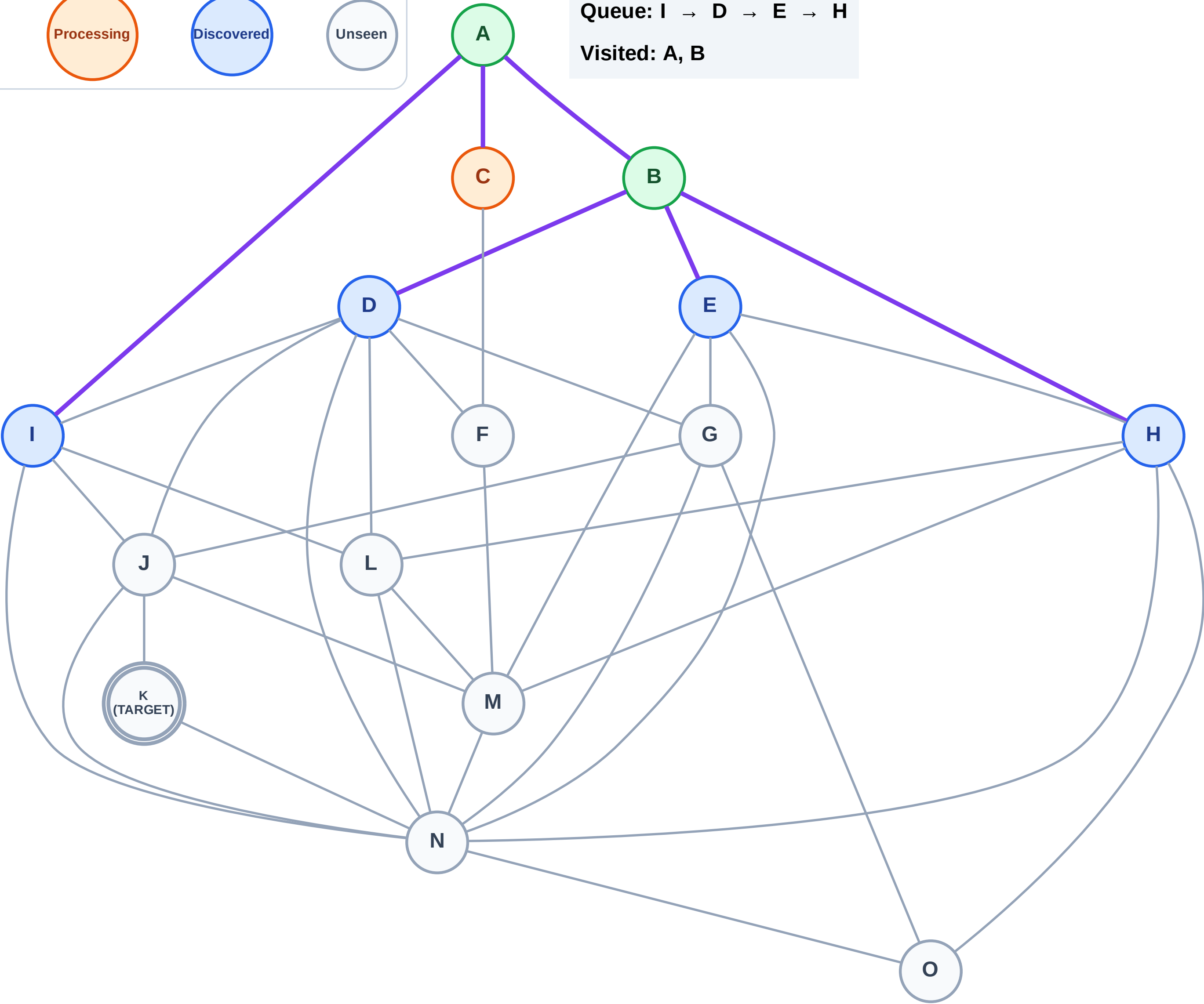
Processing

Discovered

Unseen

Queue: I → D → E → H

Visited: A, B



BFS Traversal

Step 7: Discover F from C

Legend

Complete

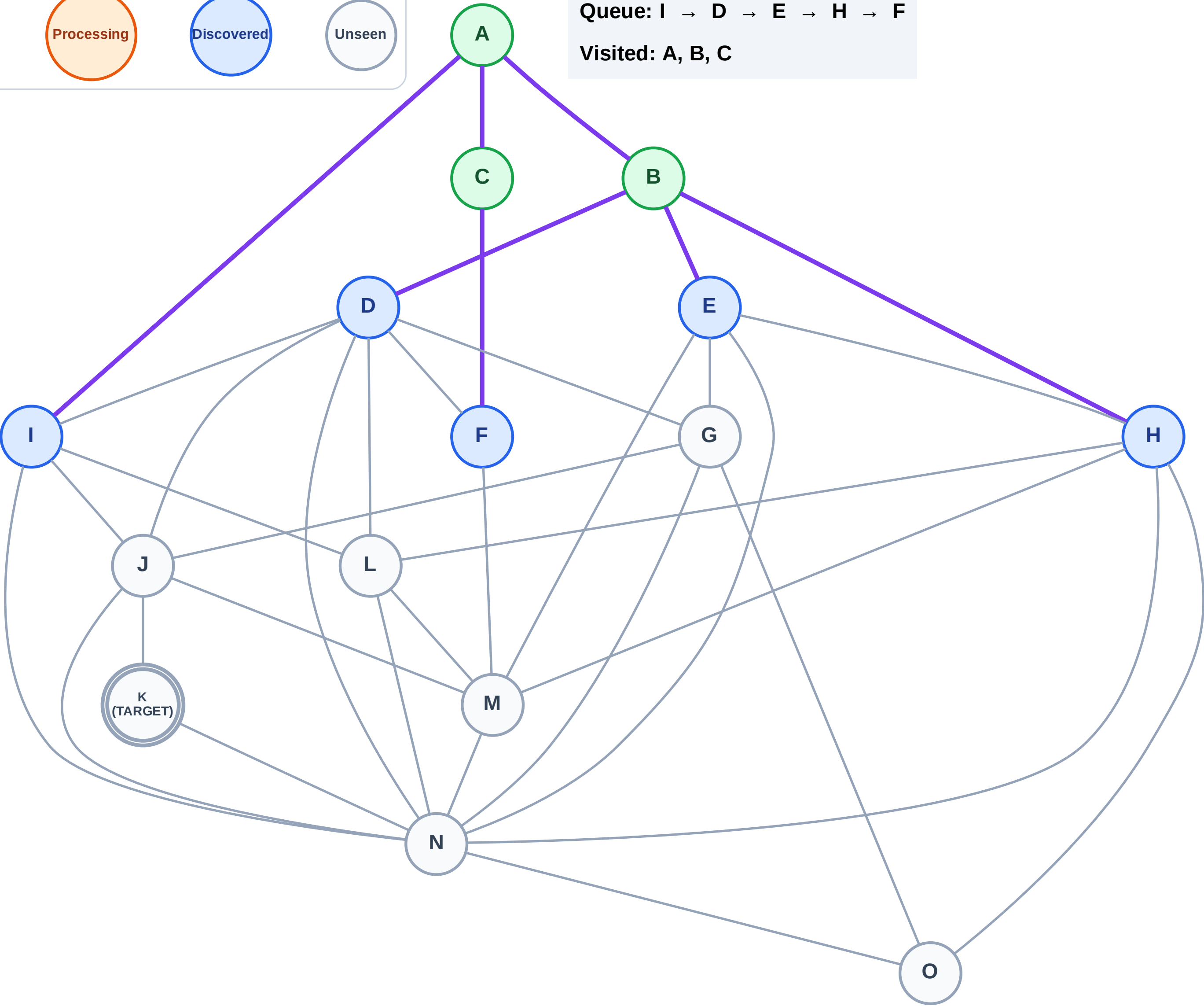
Processing

Discovered

Unseen

Queue: I → D → E → H → F

Visited: A, B, C



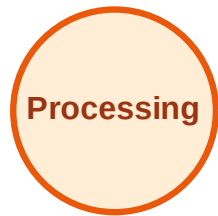
BFS Traversal

Step 8: Process node I

Legend



Complete



Processing



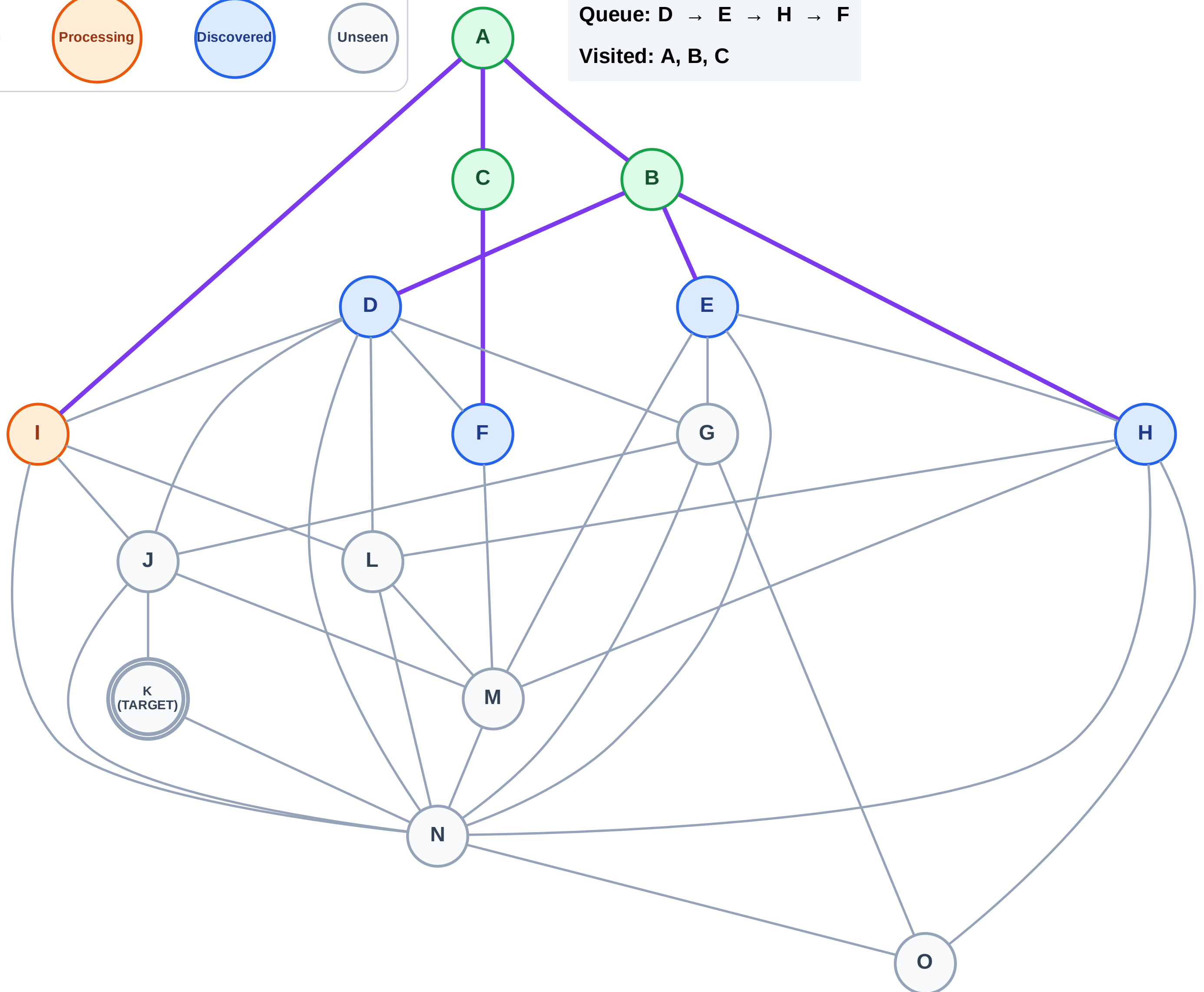
Discovered



Unseen

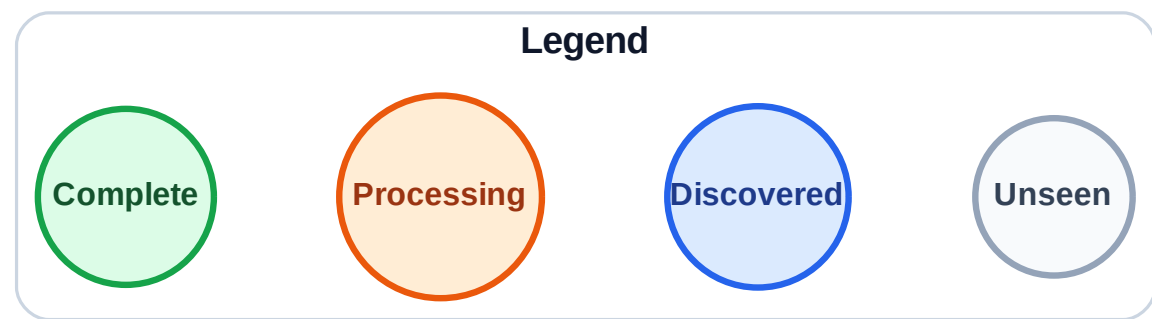
Queue: D → E → H → F

Visited: A, B, C



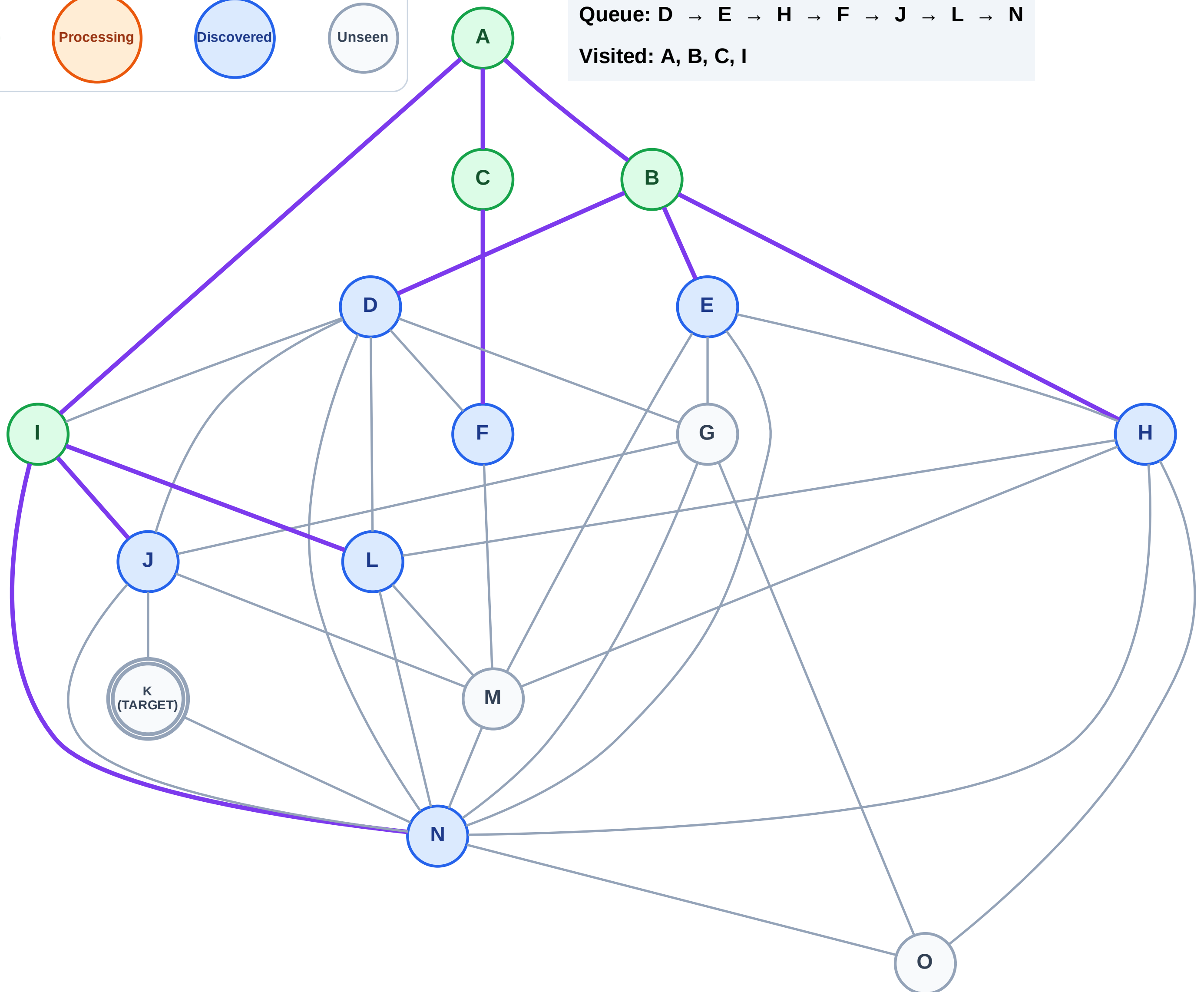
BFS Traversal

Step 9: Discover J, L, N from I



Queue: D → E → H → F → J → L → N

Visited: A, B, C, I



BFS Traversal

Step 10: Process node D

Legend

Complete

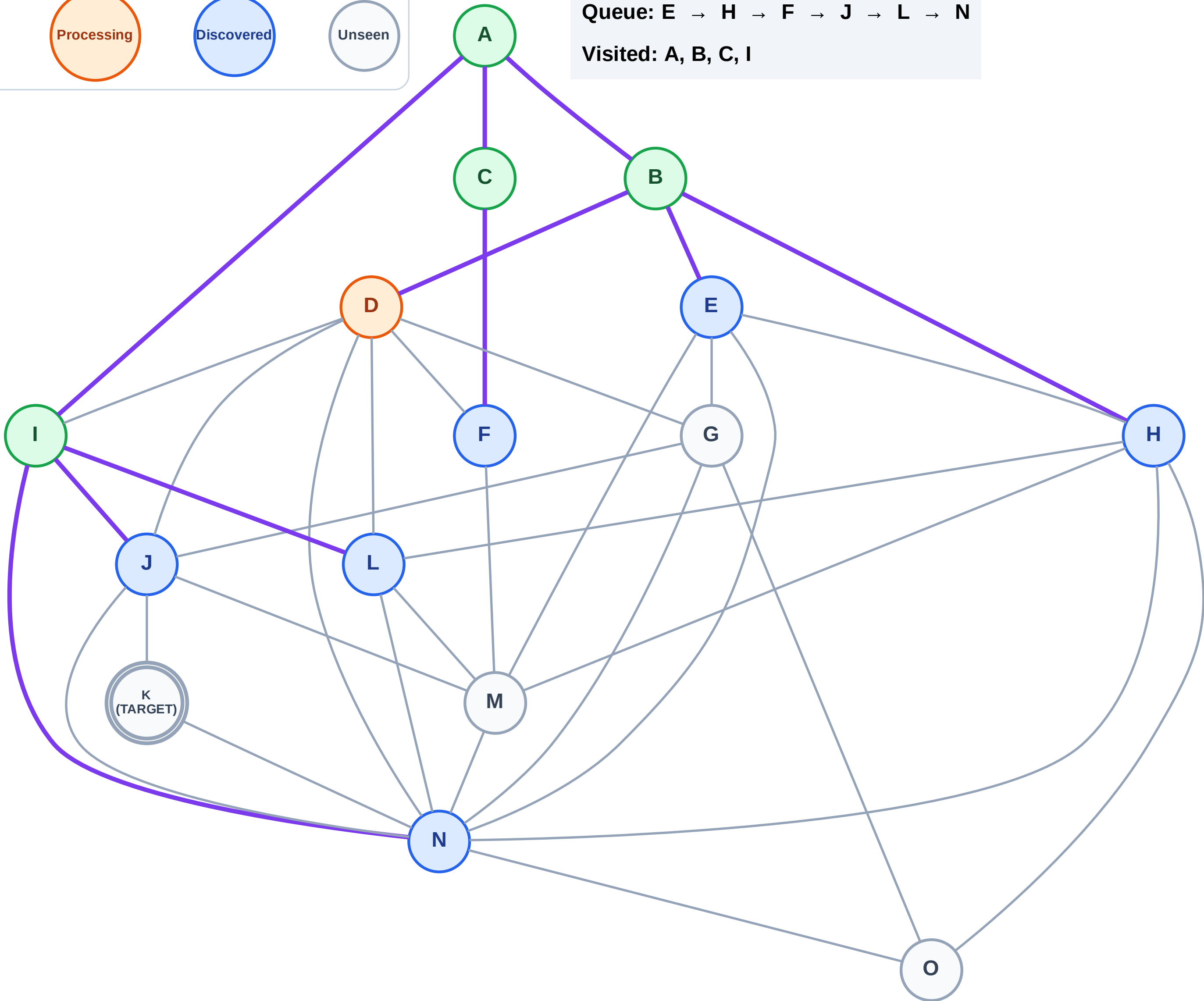
Processing

Discovered

Unseen

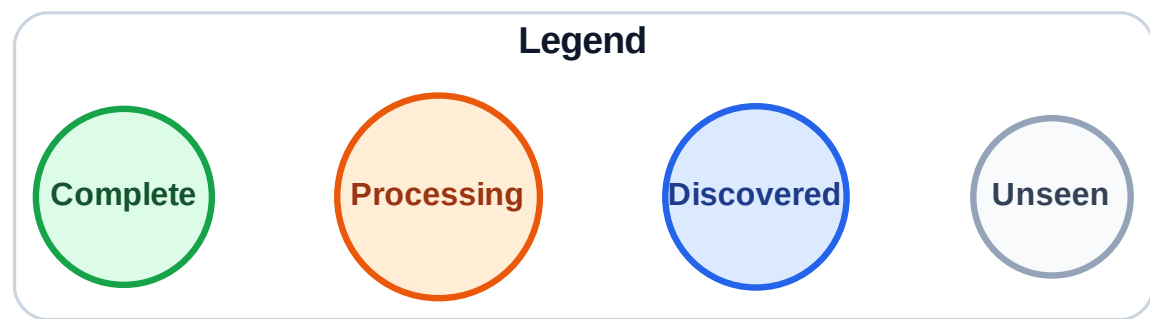
Queue: E → H → F → J → L → N

Visited: A, B, C, I



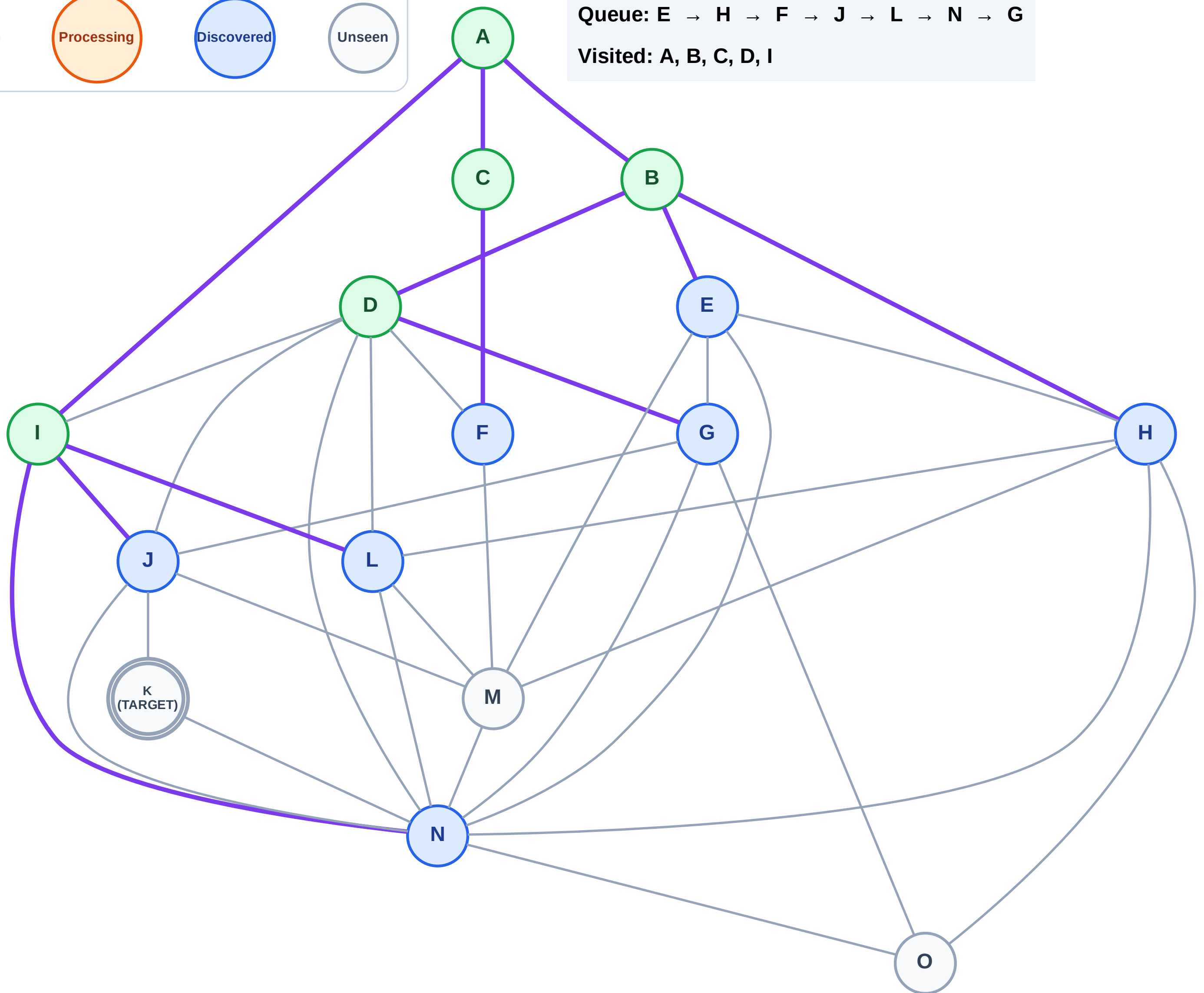
BFS Traversal

Step 11: Discover G from D



Queue: E → H → F → J → L → N → G

Visited: A, B, C, D, I



BFS Traversal

Step 12: Process node E

Legend

Complete

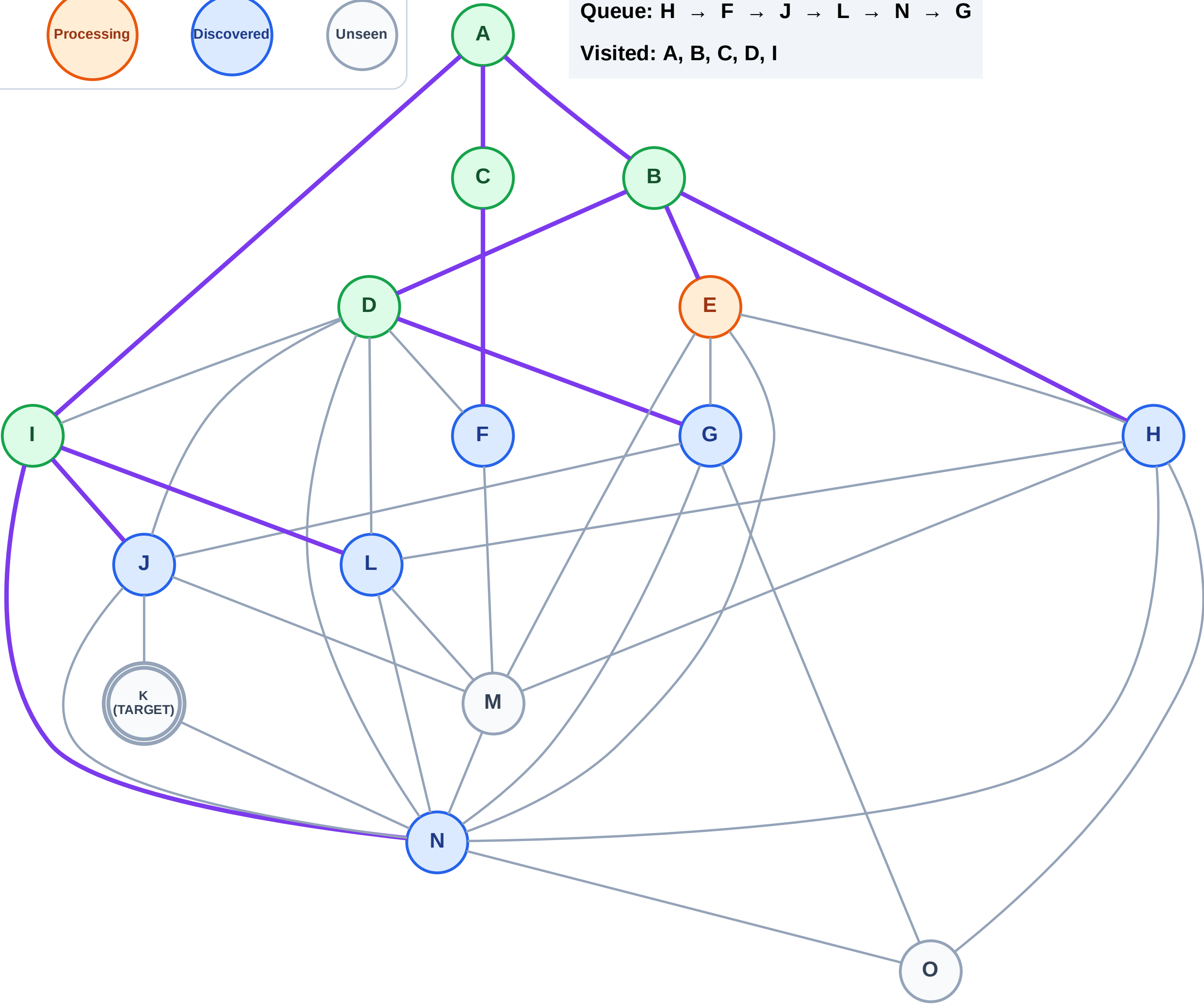
Processing

Discovered

Unseen

Queue: H → F → J → L → N → G

Visited: A, B, C, D, I



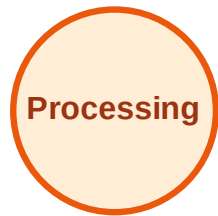
BFS Traversal

Step 13: Discover M from E

Legend



Complete



Processing



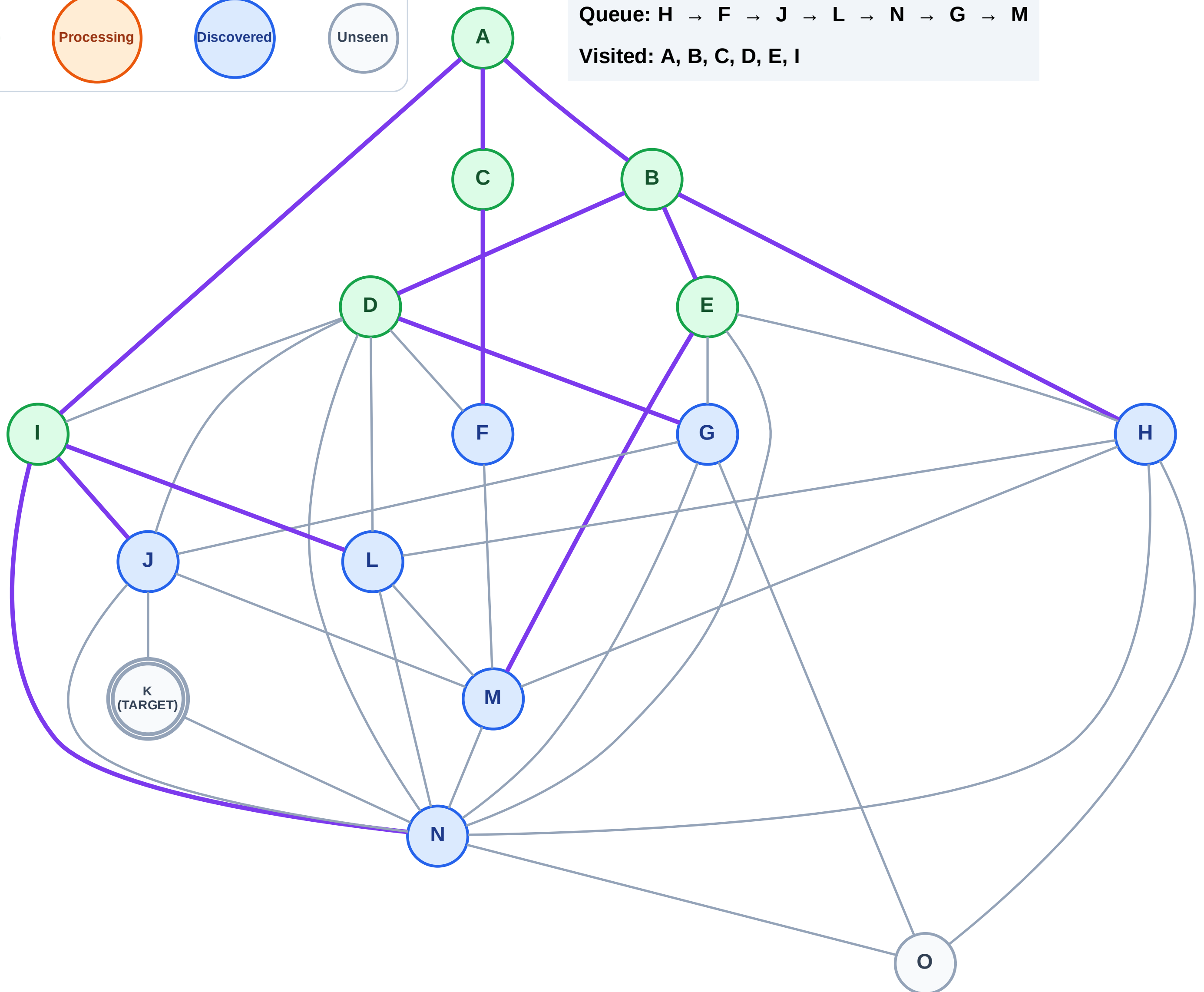
Discovered



Unseen

Queue: H → F → J → L → N → G → M

Visited: A, B, C, D, E, I



BFS Traversal

Step 14: Process node H

Legend

Complete

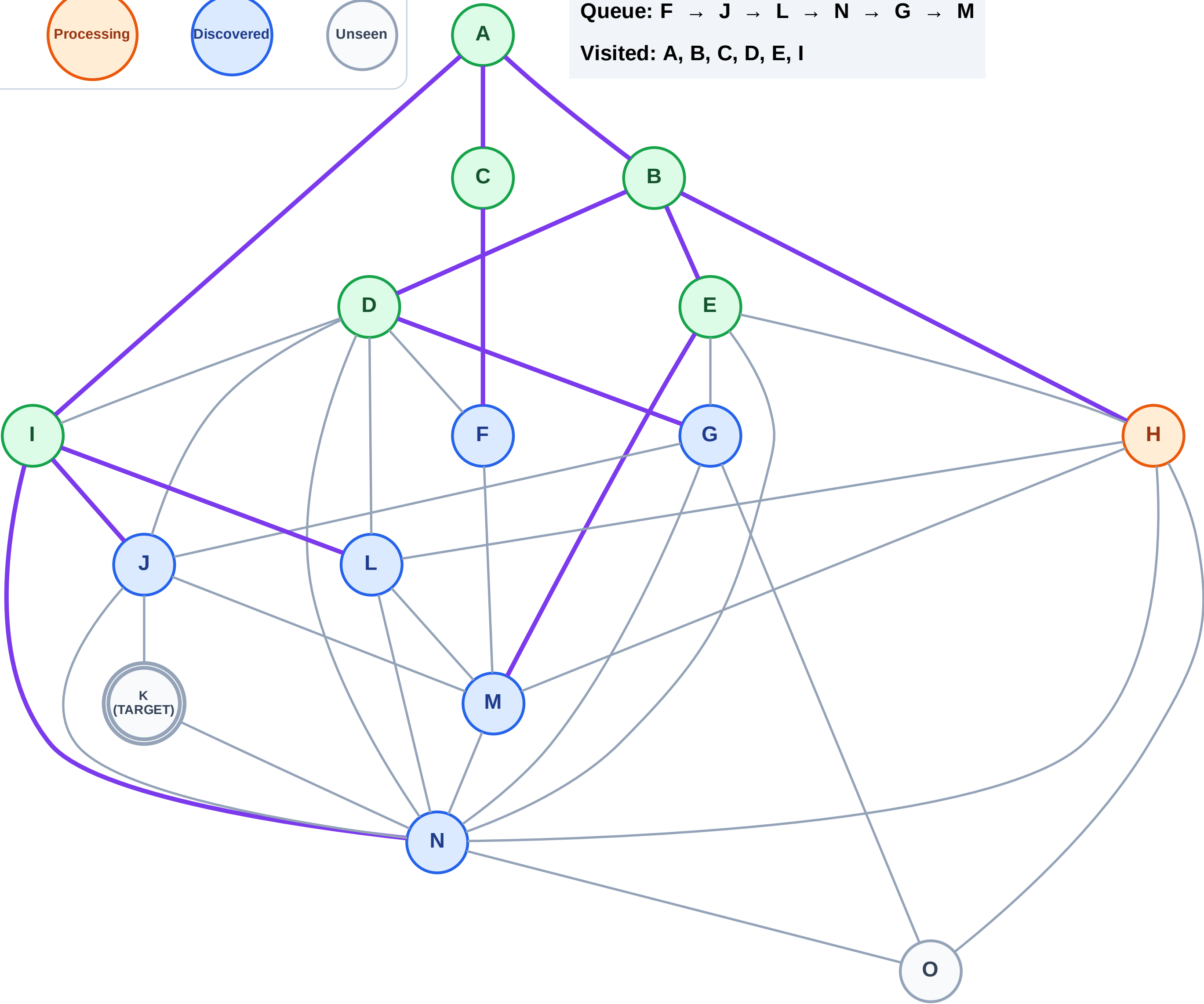
Processing

Discovered

Unseen

Queue: F → J → L → N → G → M

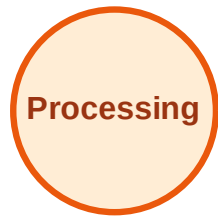
Visited: A, B, C, D, E, I



BFS Traversal

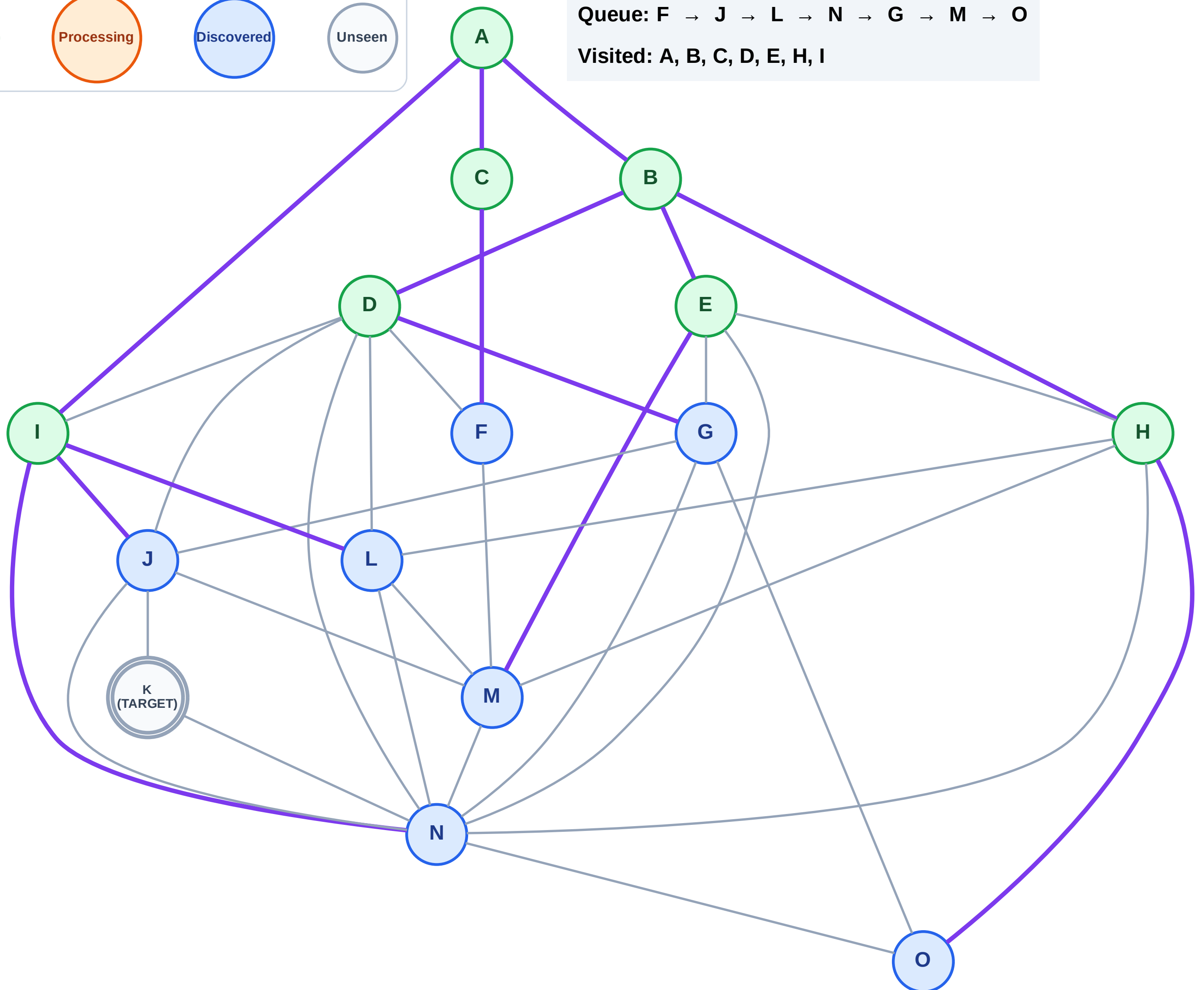
Step 15: Discover O from H

Legend



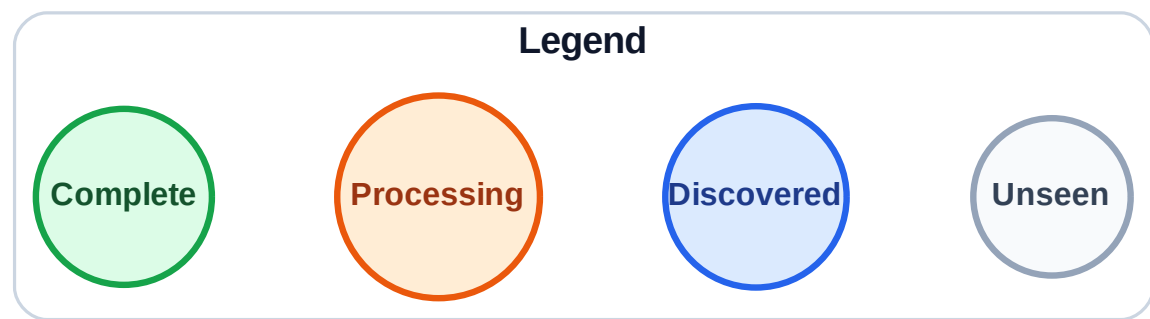
Queue: F → J → L → N → G → M → O

Visited: A, B, C, D, E, H, I



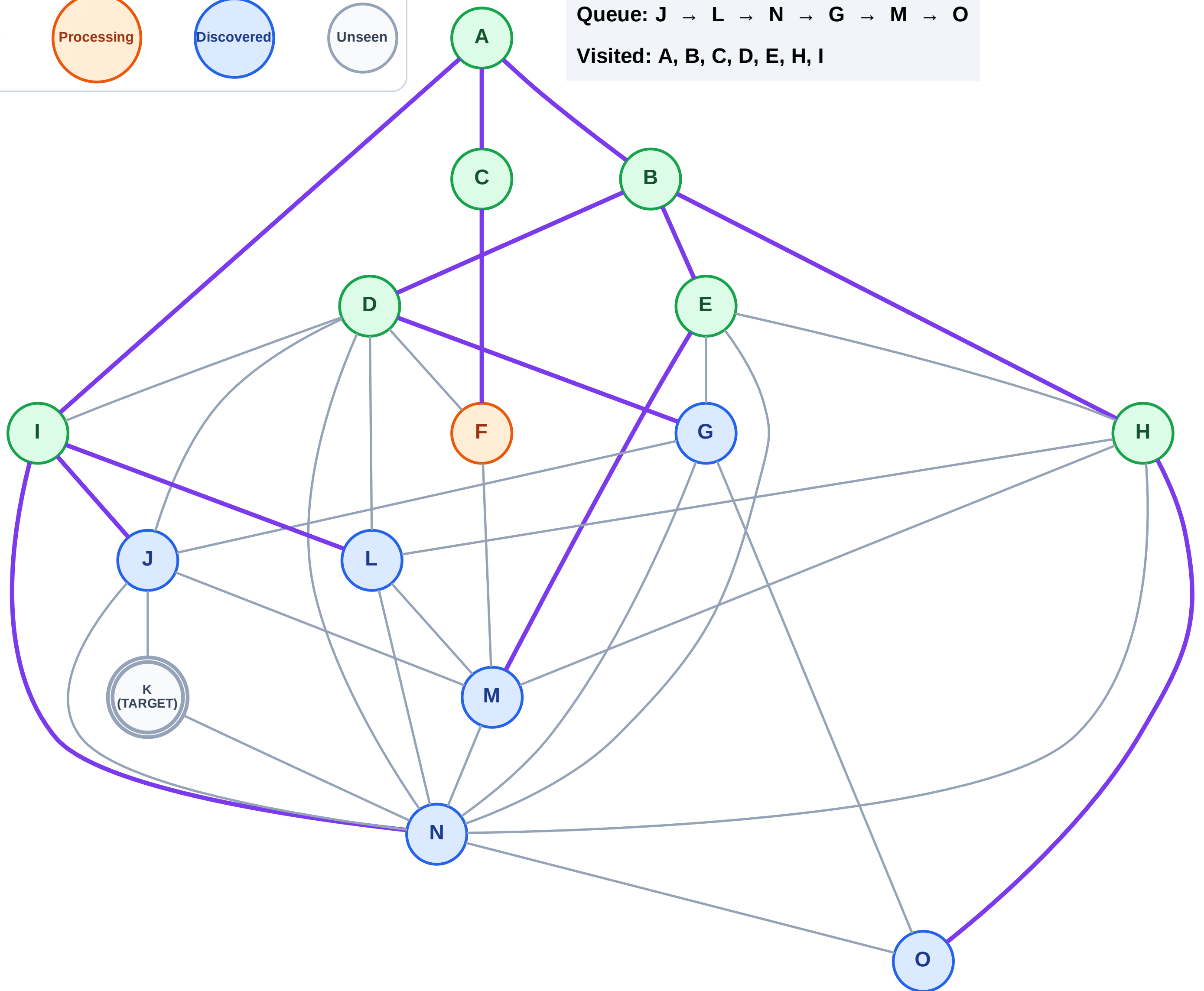
BFS Traversal

Step 16: Process node F



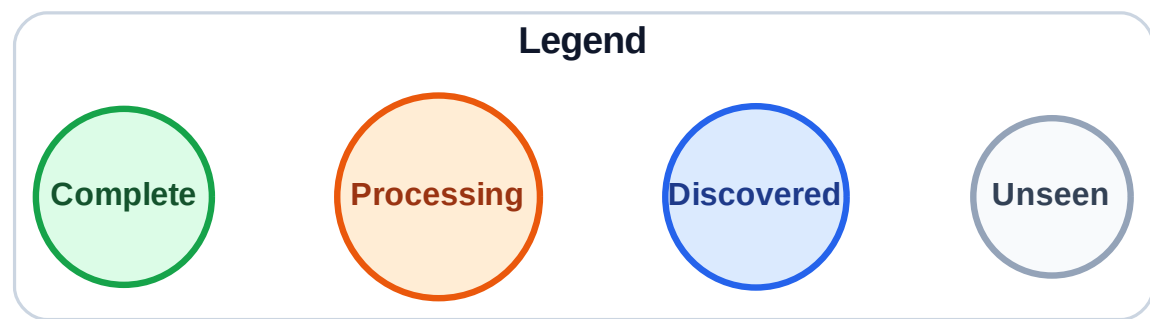
Queue: J → L → N → G → M → O

Visited: A, B, C, D, E, H, I



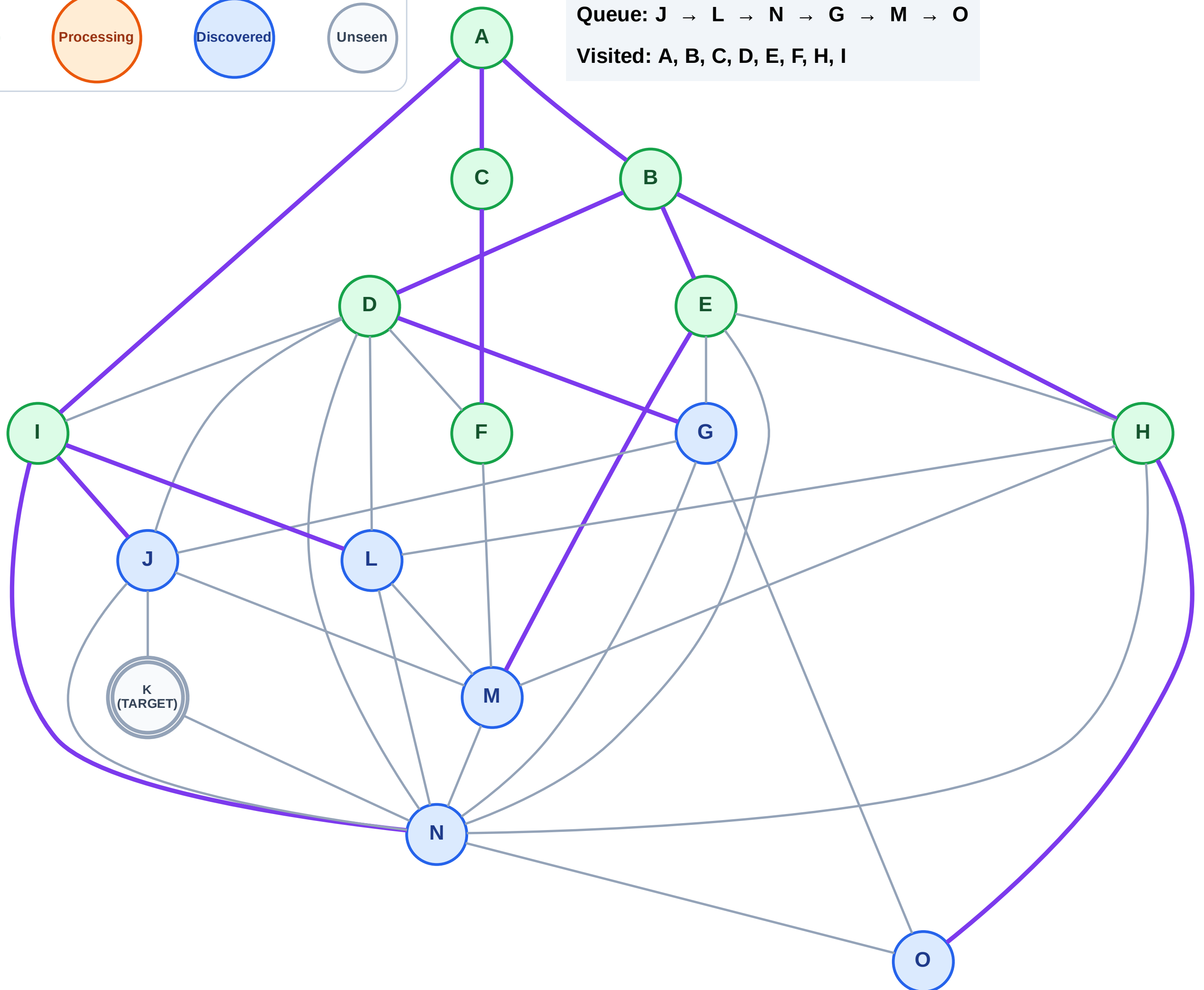
BFS Traversal

Step 17: No new neighbors from F



Queue: J → L → N → G → M → O

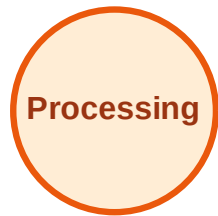
Visited: A, B, C, D, E, F, H, I



BFS Traversal

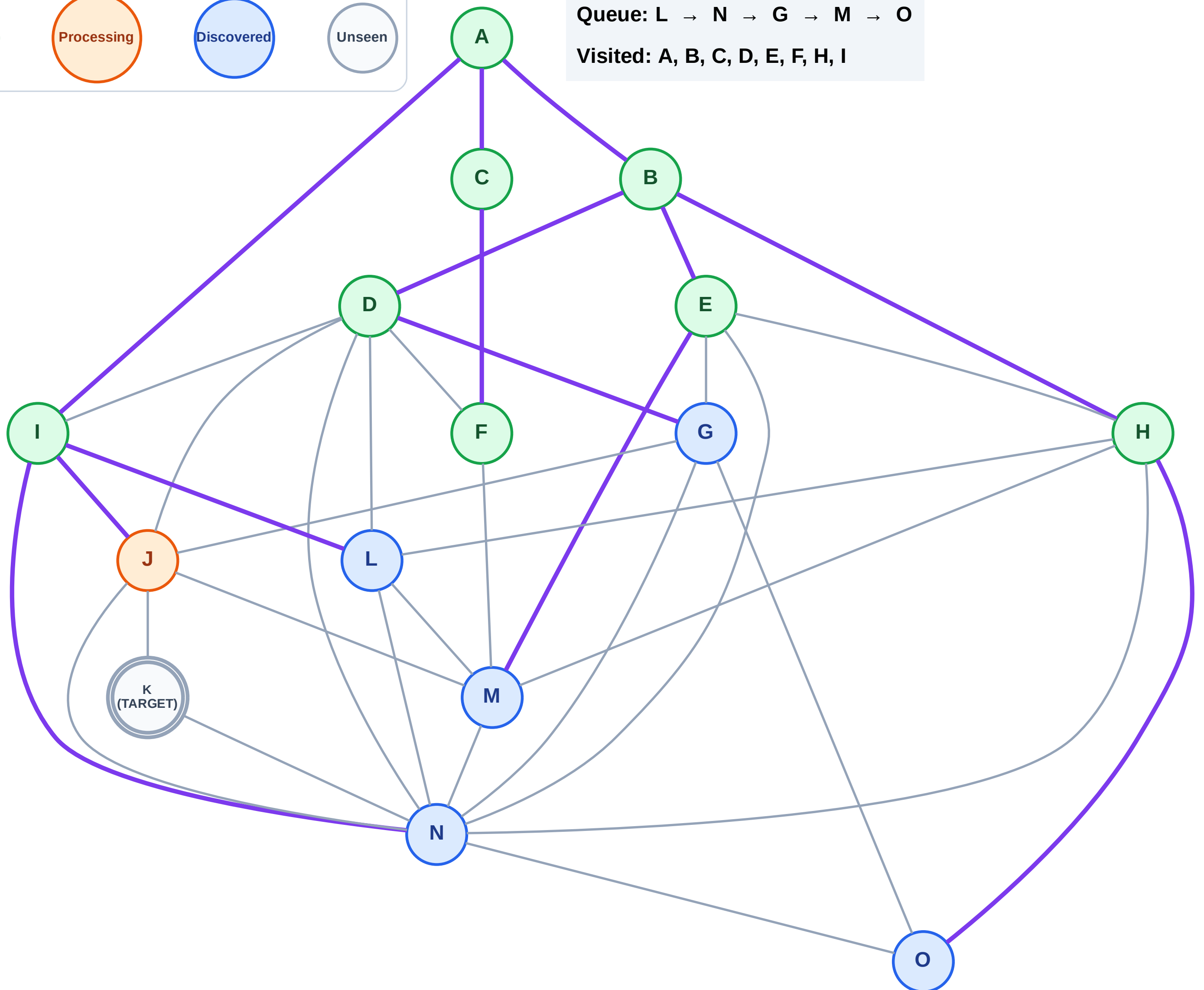
Step 18: Process node J

Legend



Queue: L → N → G → M → O

Visited: A, B, C, D, E, F, H, I



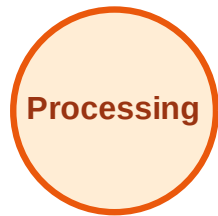
BFS Traversal

Step 19: Discover K from J

Legend



Complete



Processing



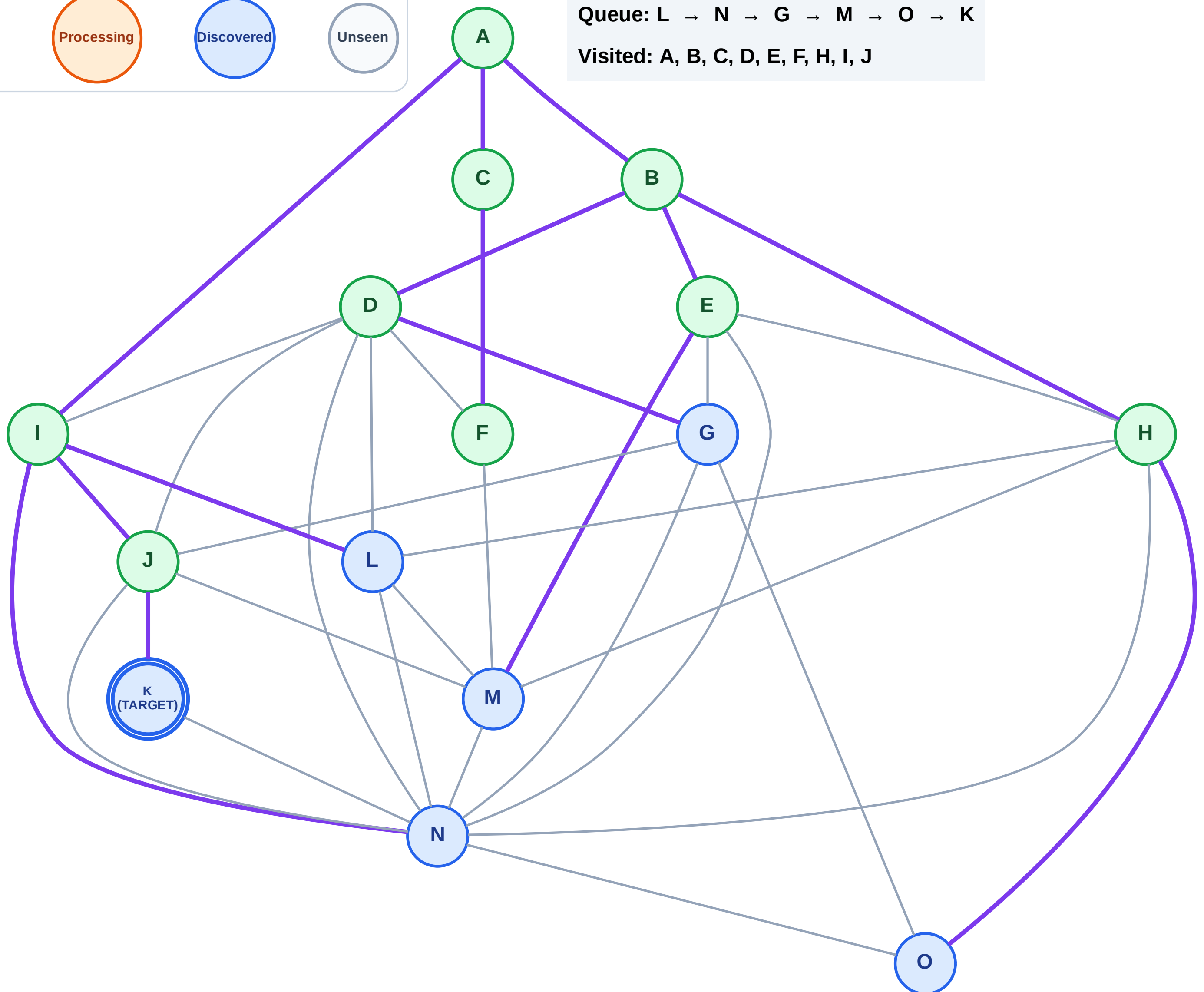
Discovered



Unseen

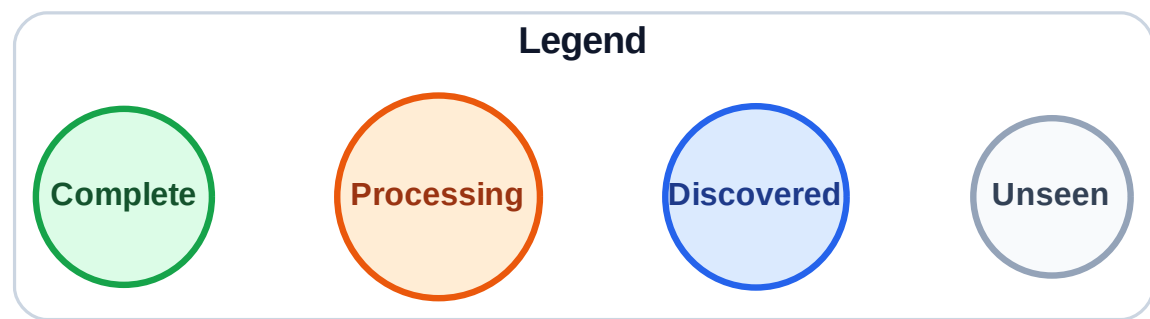
Queue: L → N → G → M → O → K

Visited: A, B, C, D, E, F, H, I, J



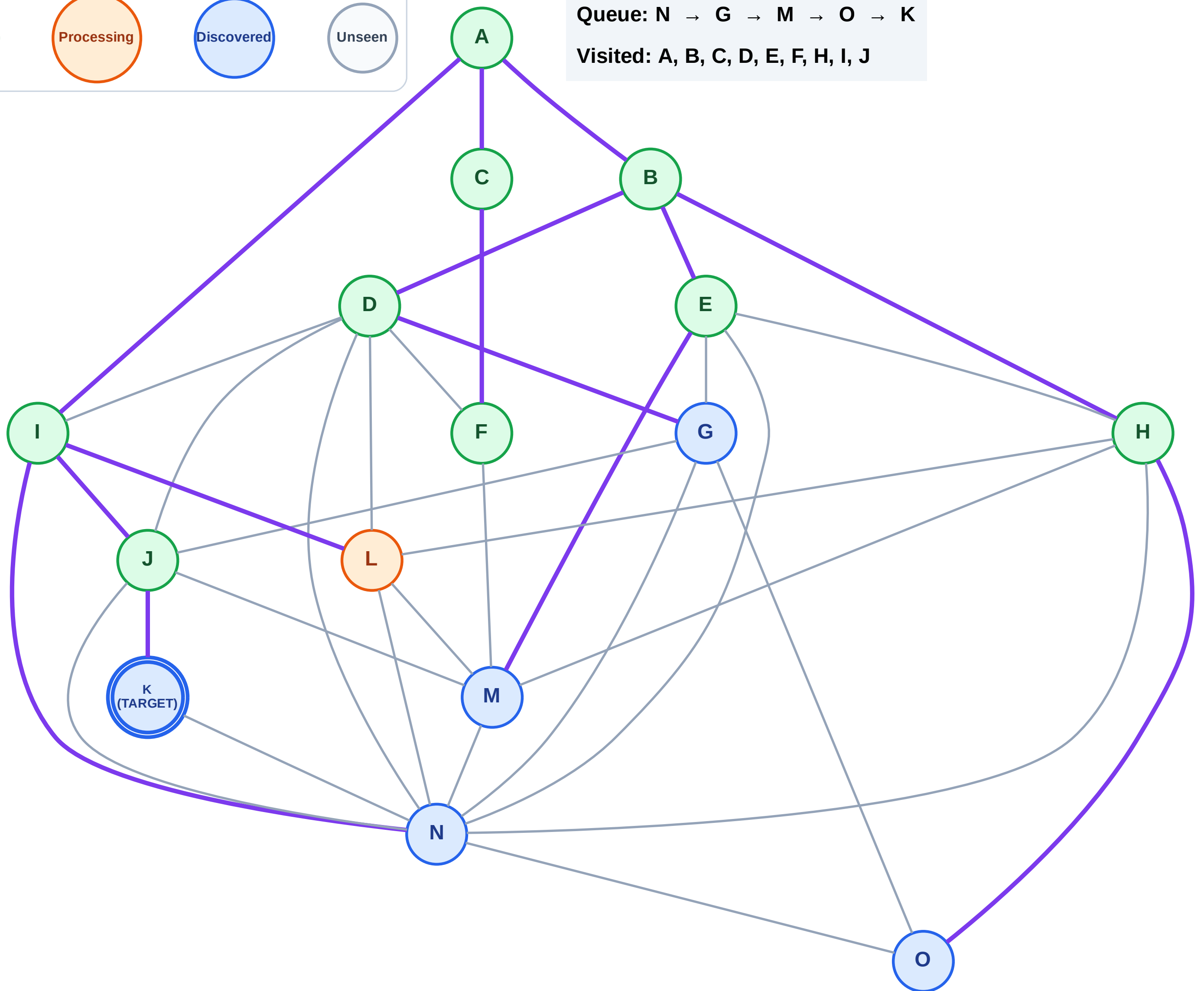
BFS Traversal

Step 20: Process node L



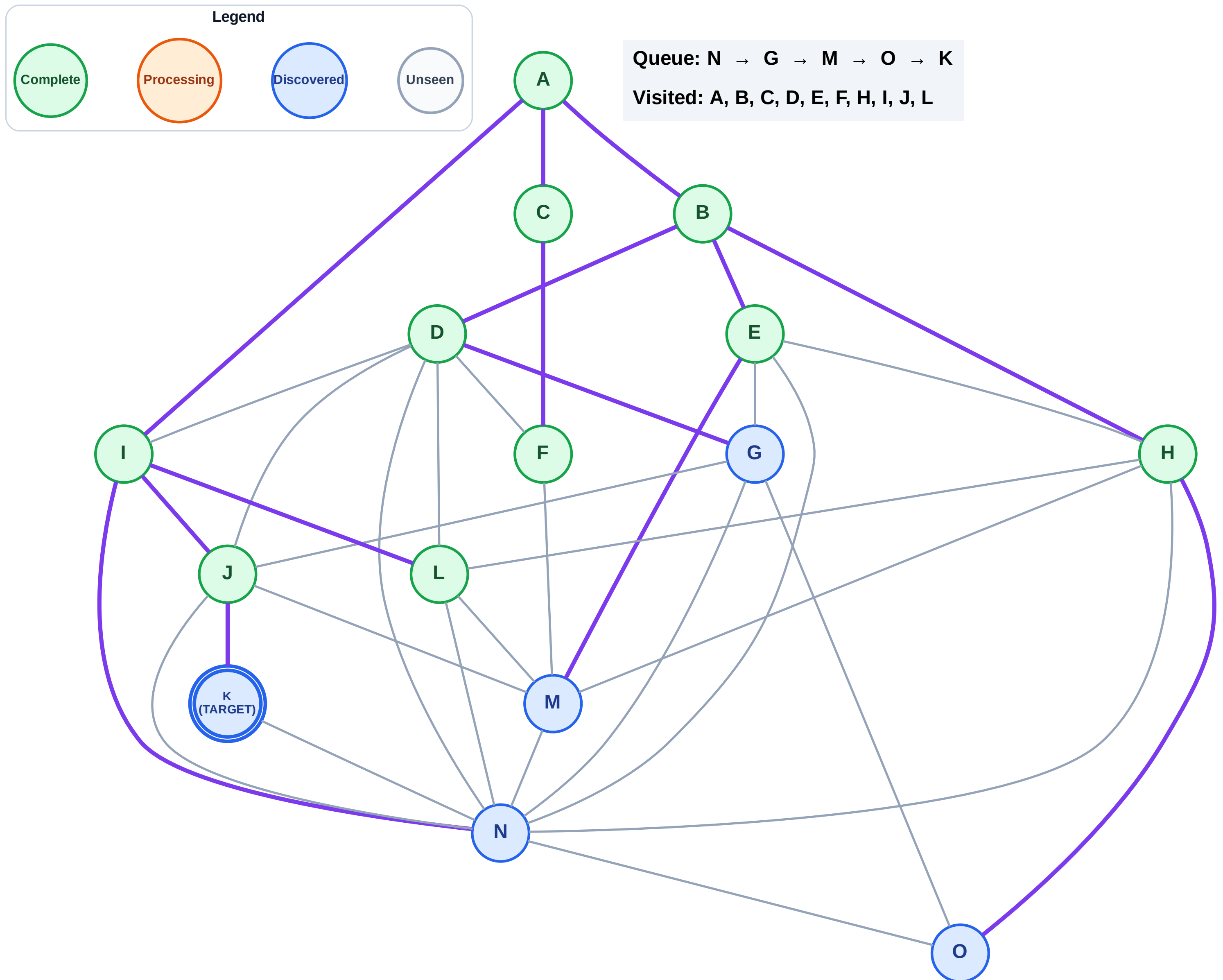
Queue: N → G → M → O → K

Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

Step 21: No new neighbors from L



BFS Traversal

Step 22: Process node N

Legend

Complete

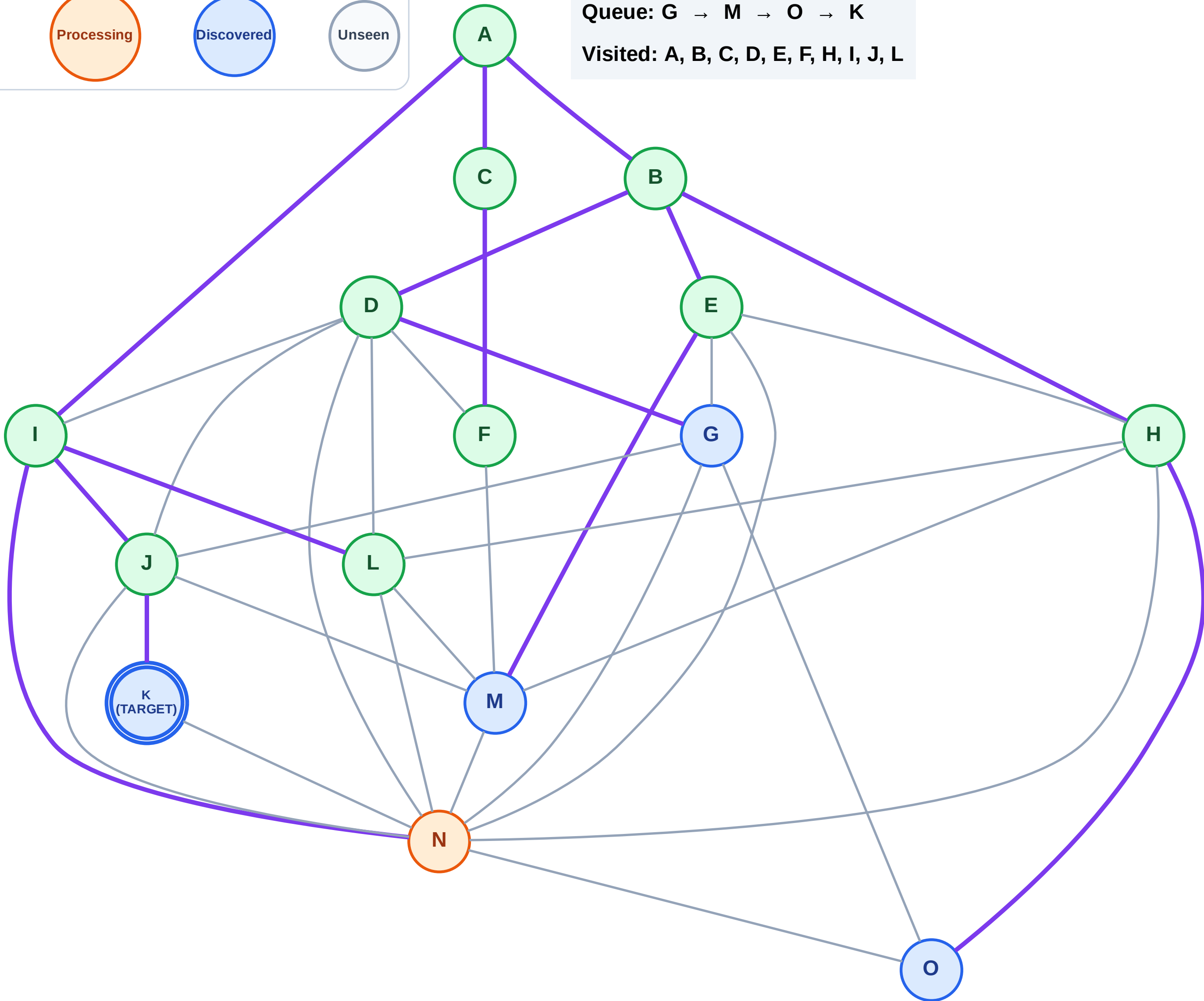
Processing

Discovered

Unseen

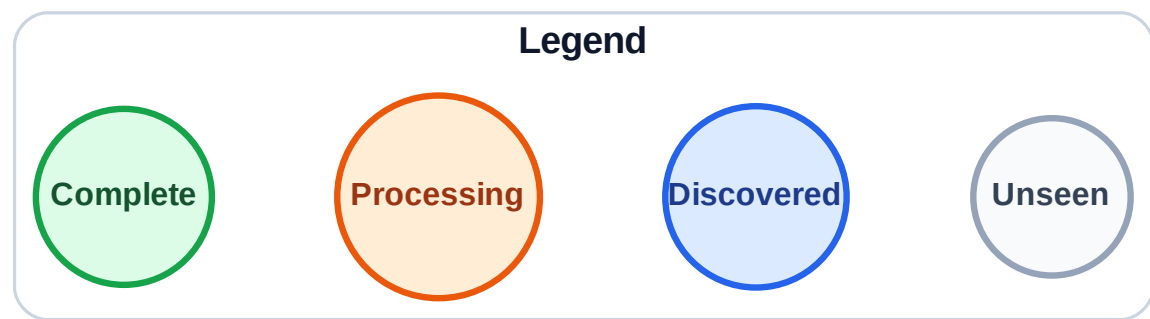
Queue: G → M → O → K

Visited: A, B, C, D, E, F, H, I, J, L



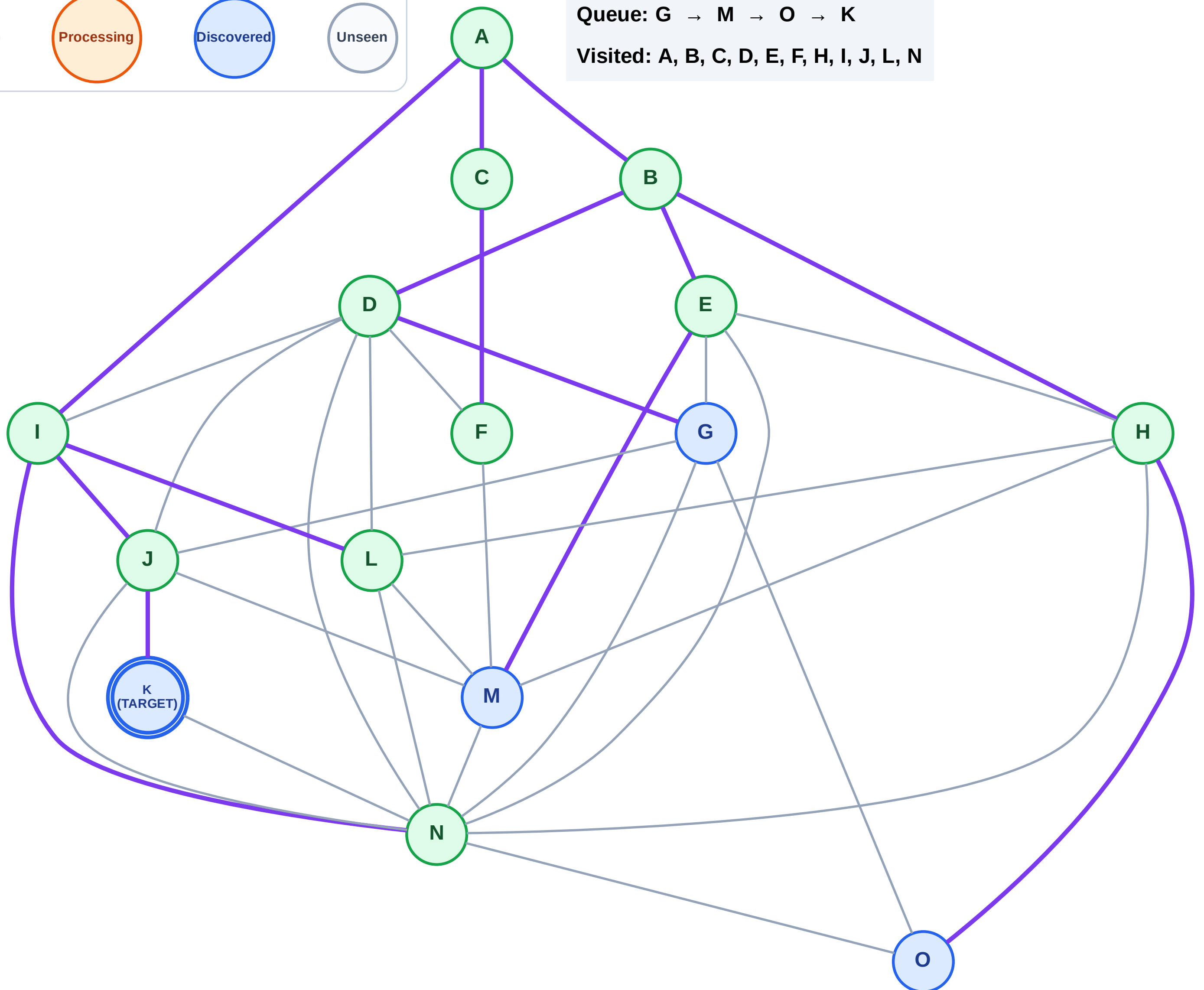
BFS Traversal

Step 23: No new neighbors from N



Queue: G → M → O → K

Visited: A, B, C, D, E, F, H, I, J, L, N



BFS Traversal

Step 24: Process node G

Legend

Complete

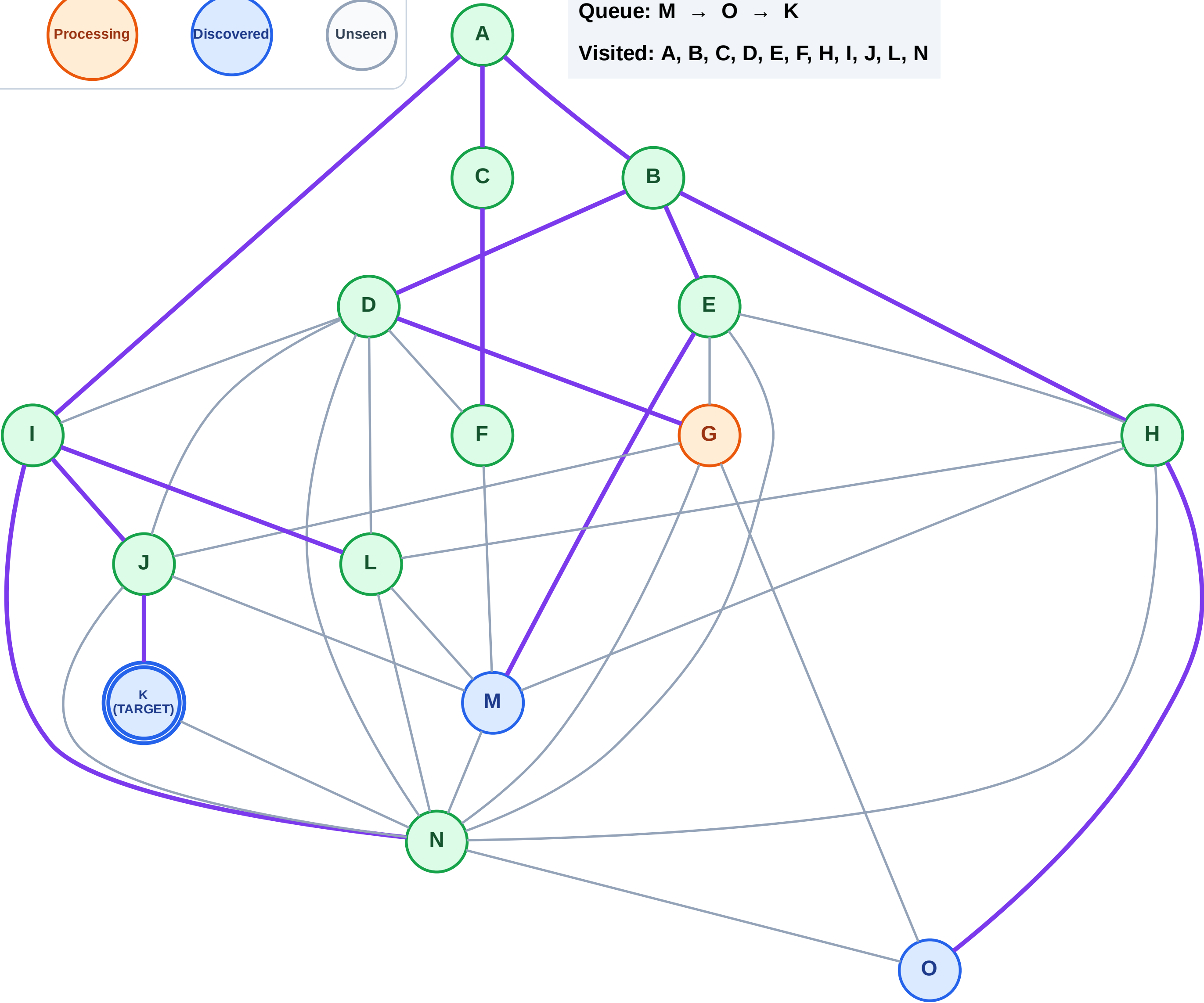
Processing

Discovered

Unseen

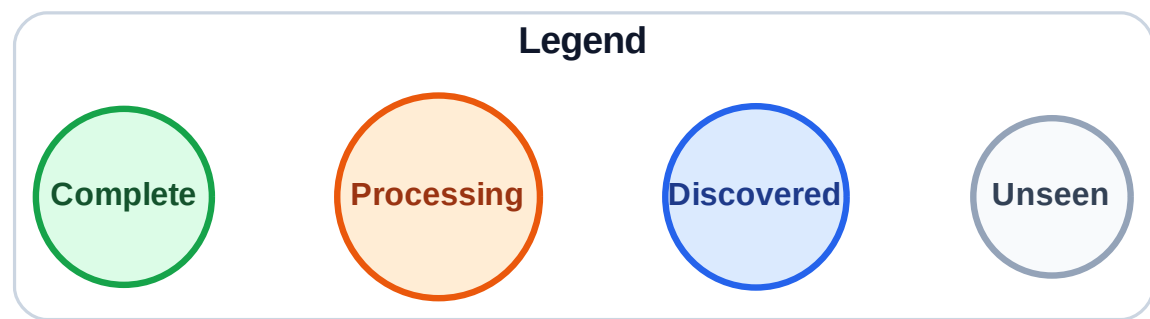
Queue: M → O → K

Visited: A, B, C, D, E, F, H, I, J, L, N



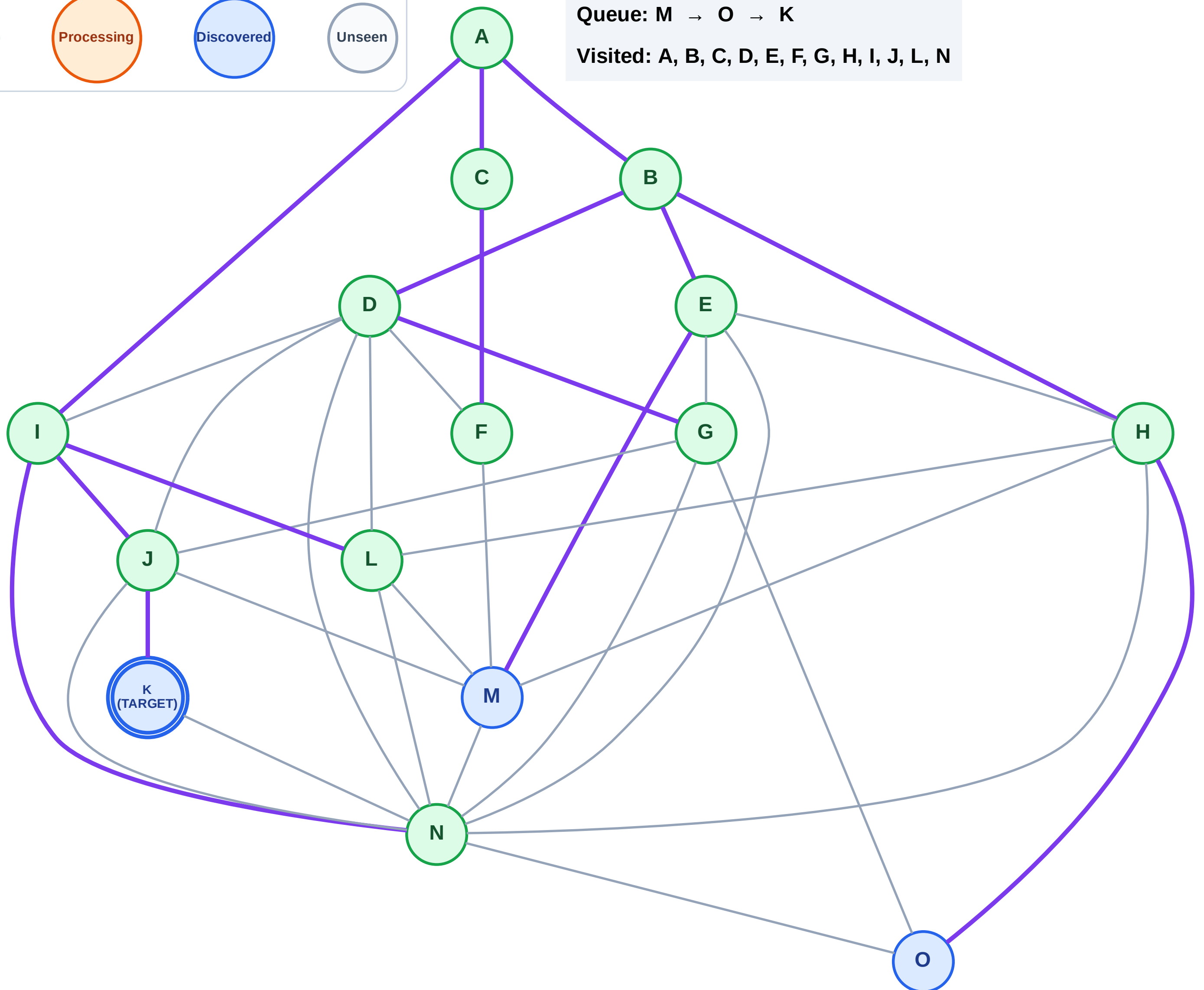
BFS Traversal

Step 25: No new neighbors from G



Queue: M → O → K

Visited: A, B, C, D, E, F, G, H, I, J, L, N



BFS Traversal

Step 26: Process node M

Legend

Complete

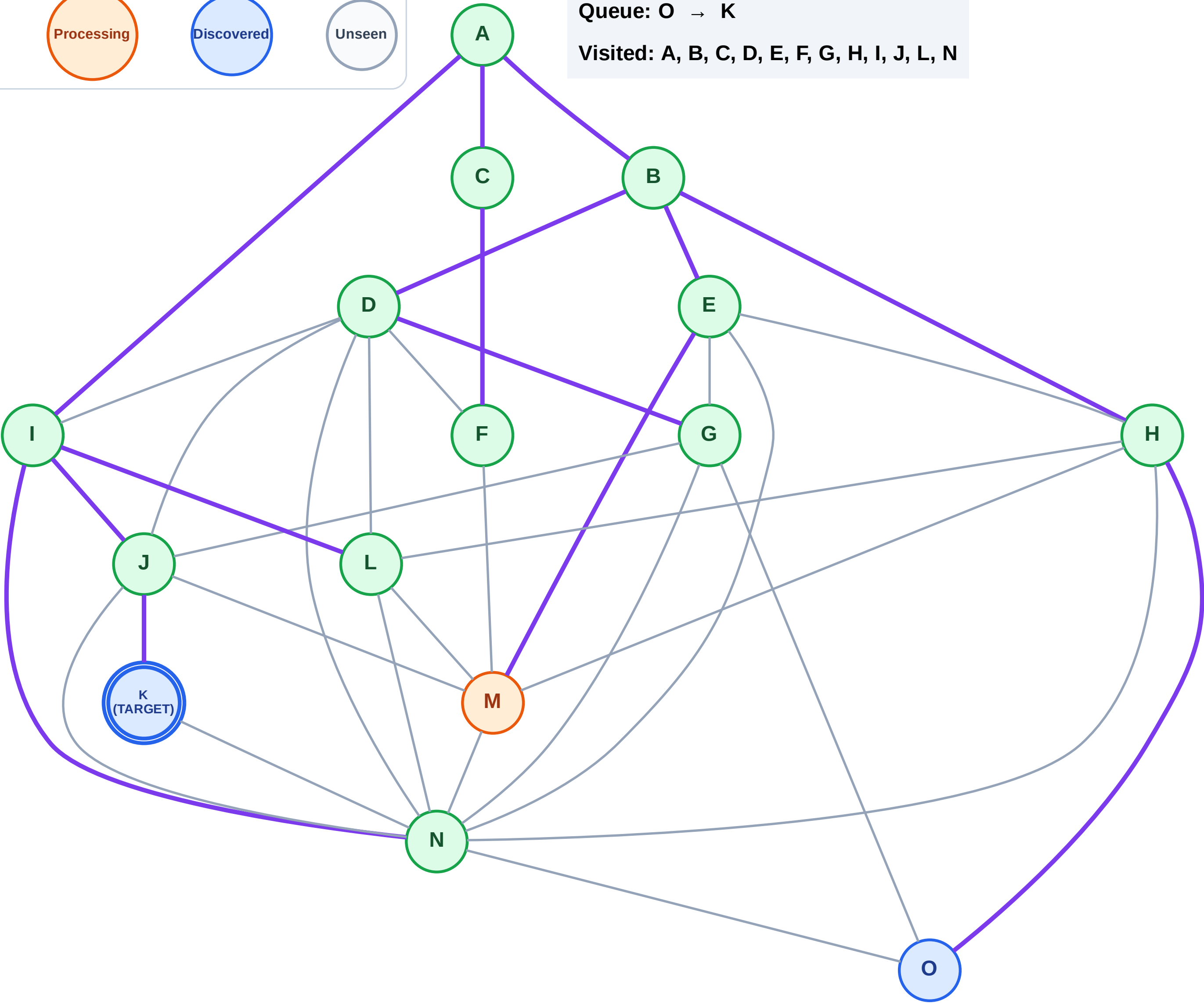
Processing

Discovered

Unseen

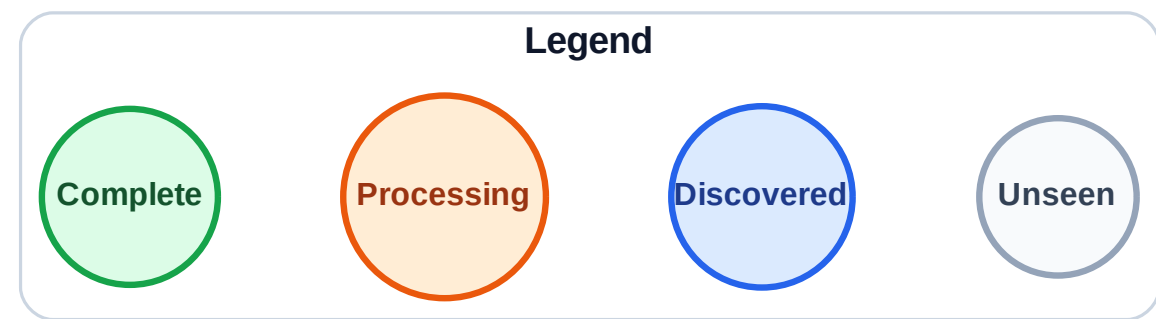
Queue: O → K

Visited: A, B, C, D, E, F, G, H, I, J, L, N



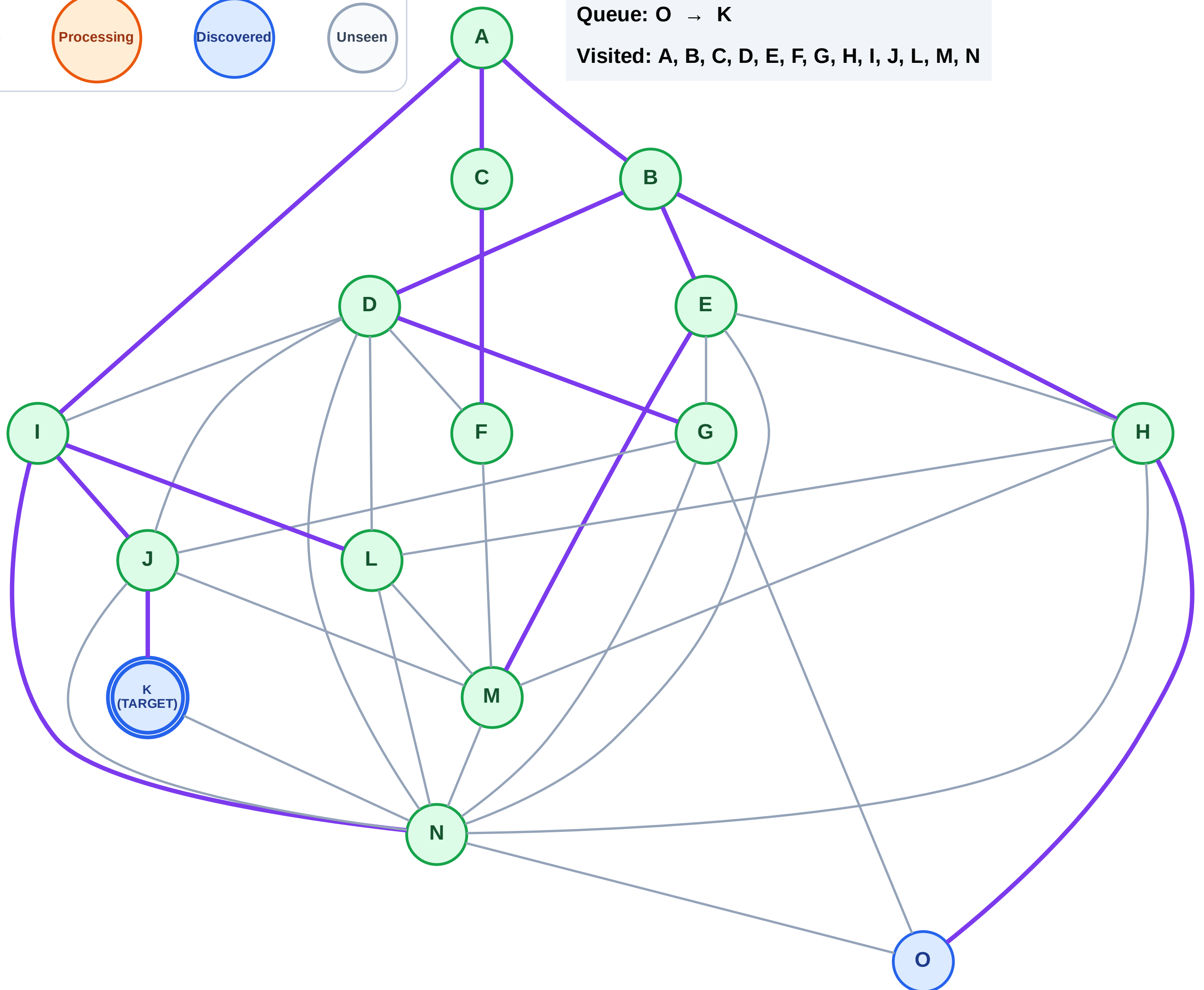
BFS Traversal

Step 27: No new neighbors from M



Queue: O → K

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N



BFS Traversal

Step 28: Process node O

Legend

Complete

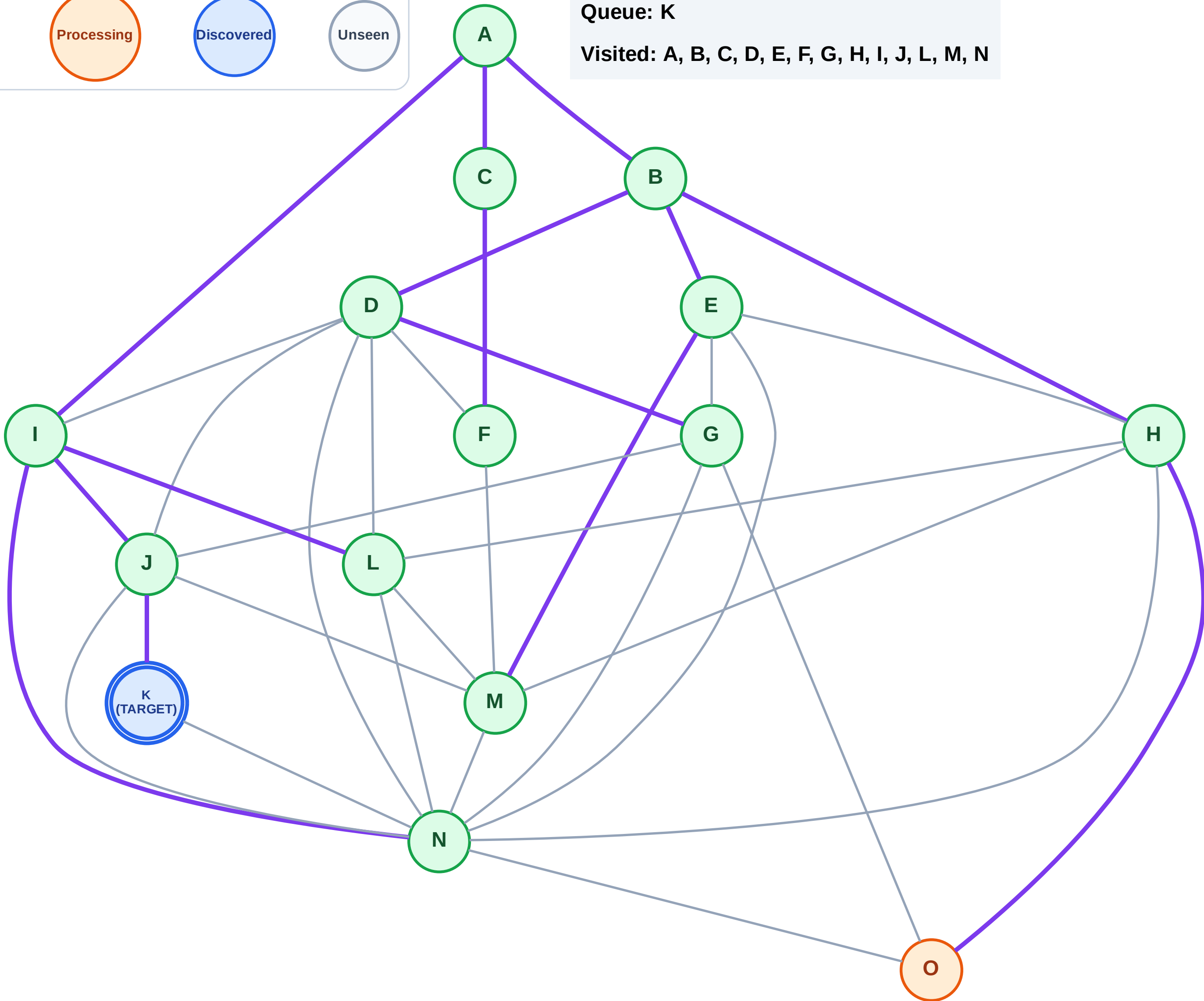
Processing

Discovered

Unseen

Queue: K

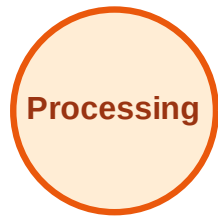
Visited: A, B, C, D, E, F, G, H, I, J, L, M, N



BFS Traversal

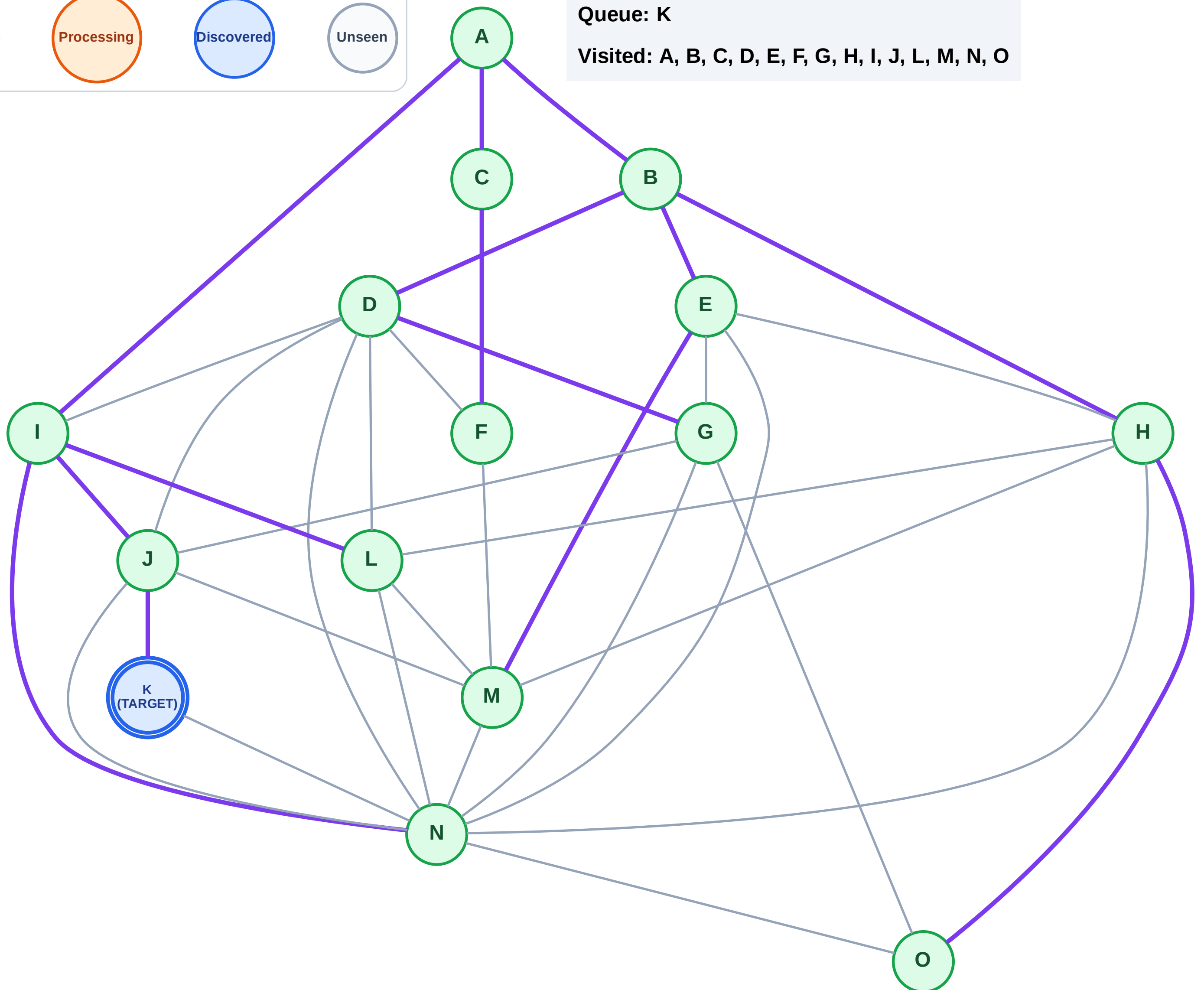
Step 29: No new neighbors from O

Legend

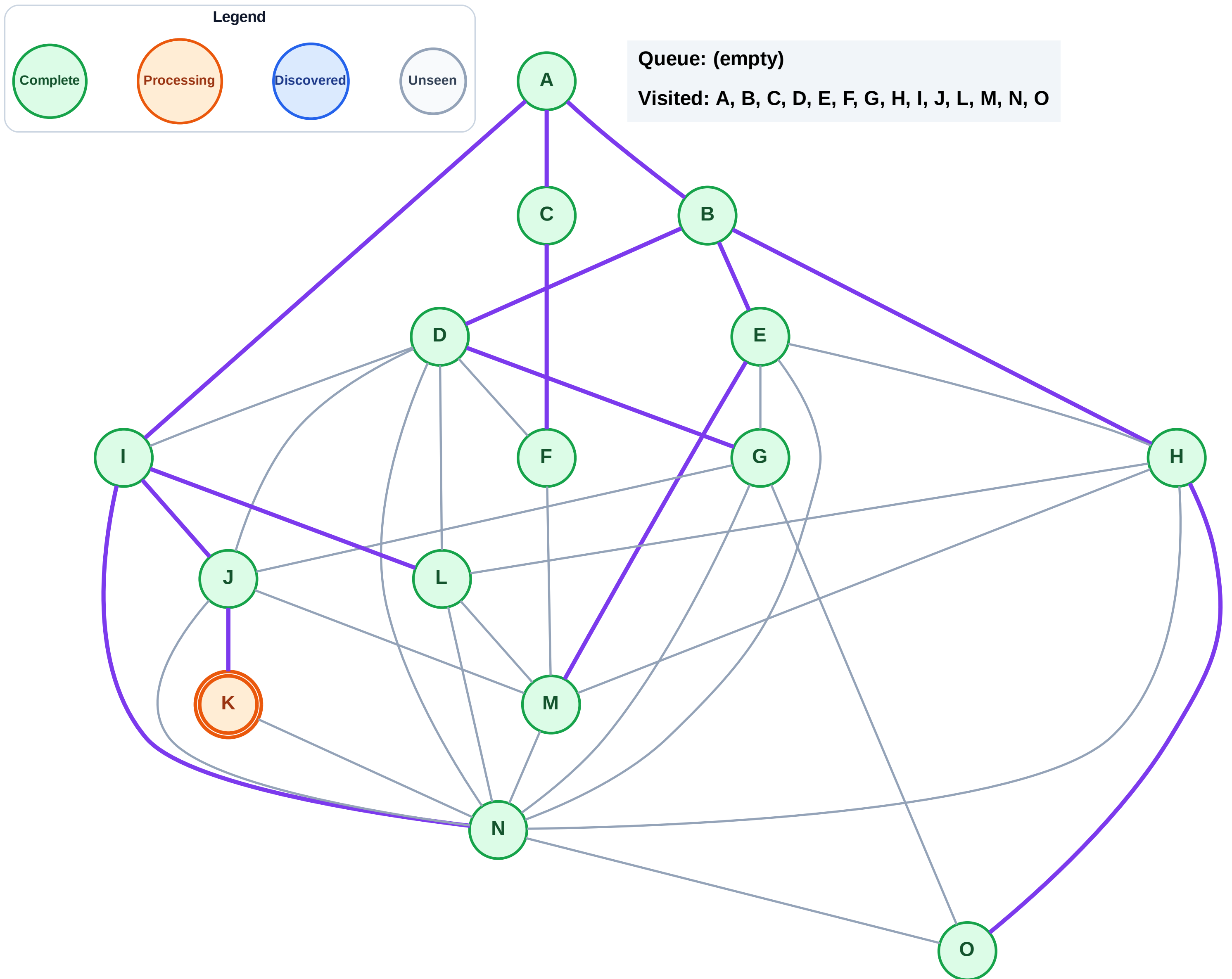


Queue: K

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N, O



Step 30: Process node K



BFS Traversal

Step 31: TARGET NODE K REACHED! Path (3 hops): A → I → J → K

Legend

Complete

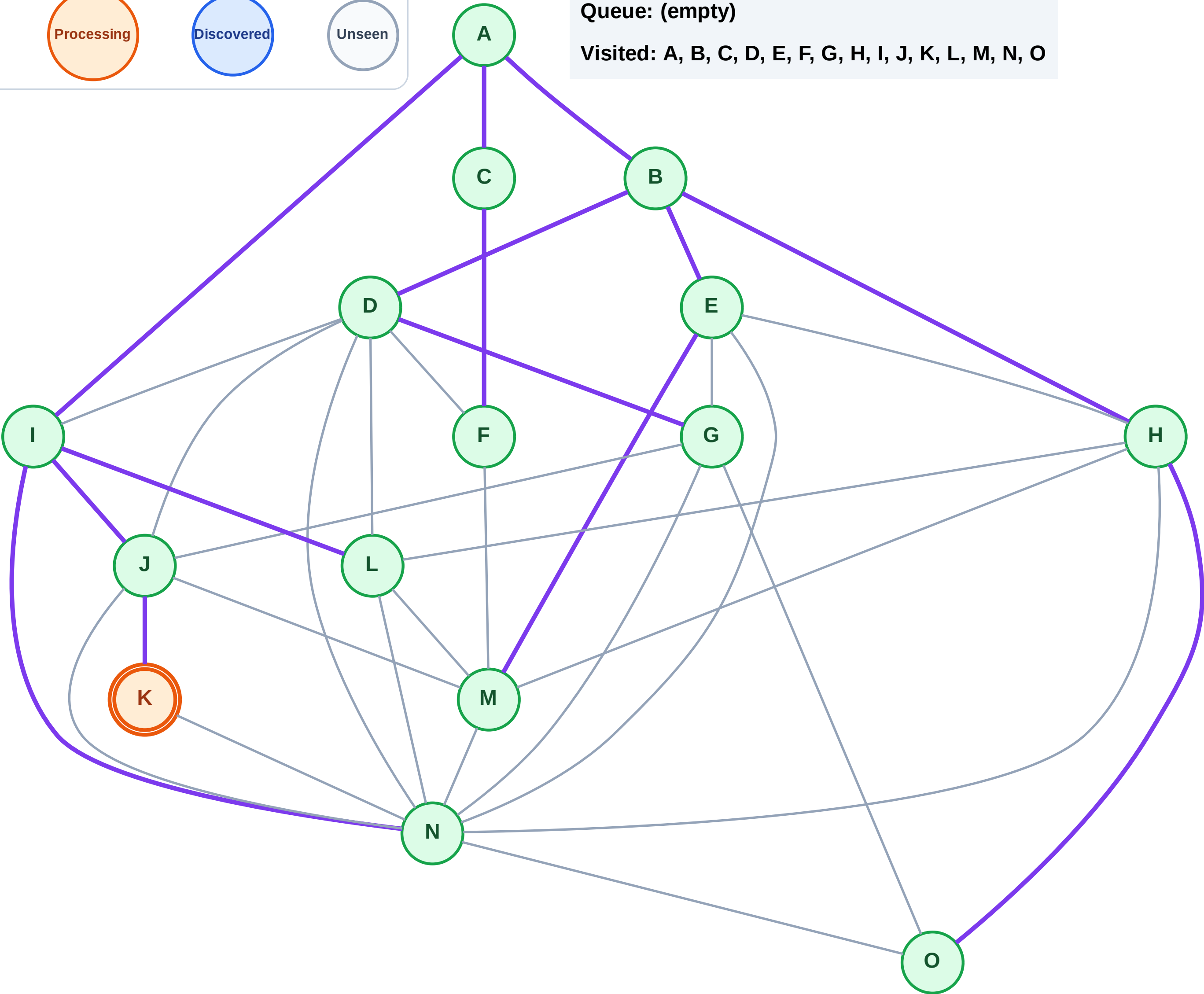
Processing

Discovered

Unseen

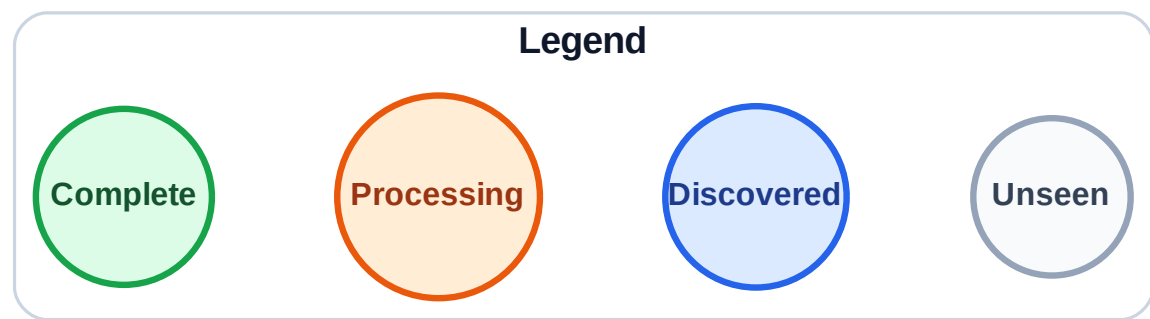
Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, N, O



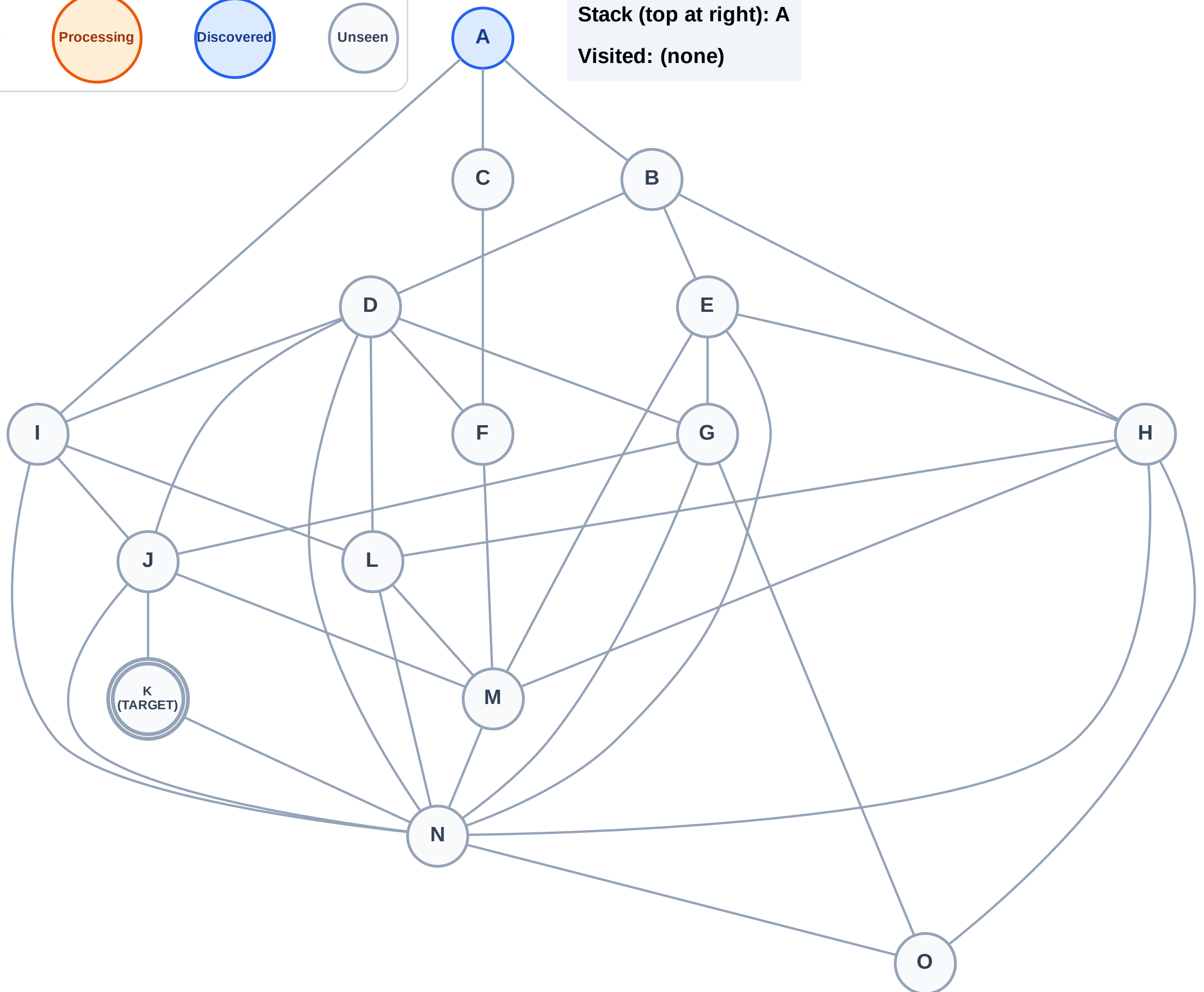
DFS Traversal

Step 1: Start at A (Seeking target node K)



Stack (top at right): A

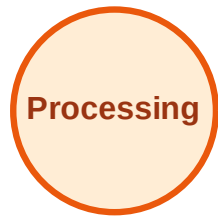
Visited: (none)



DFS Traversal

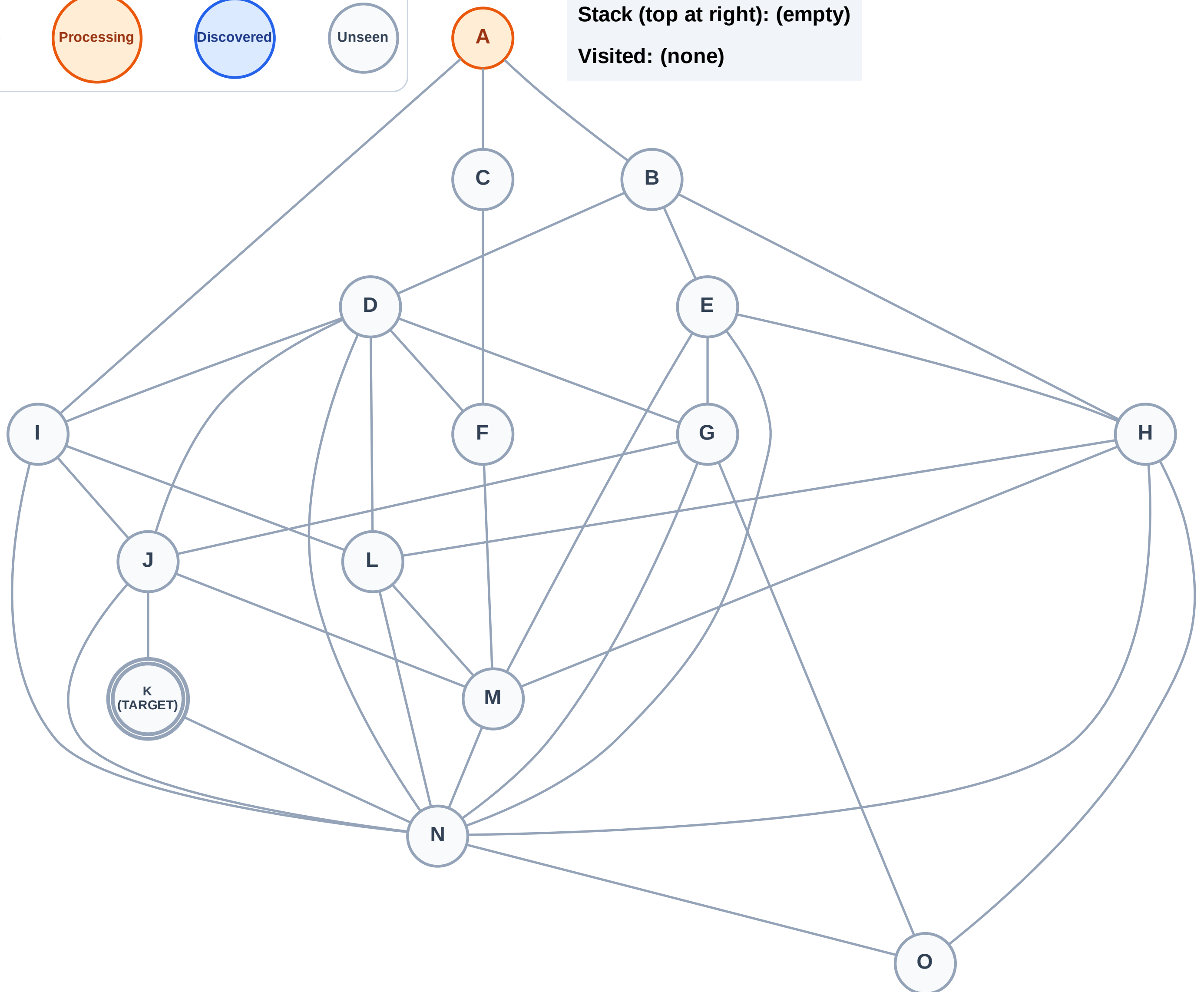
Step 2: Process node A

Legend



Stack (top at right): (empty)

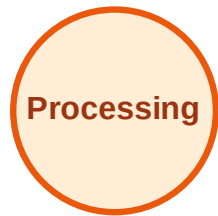
Visited: (none)



DFS Traversal

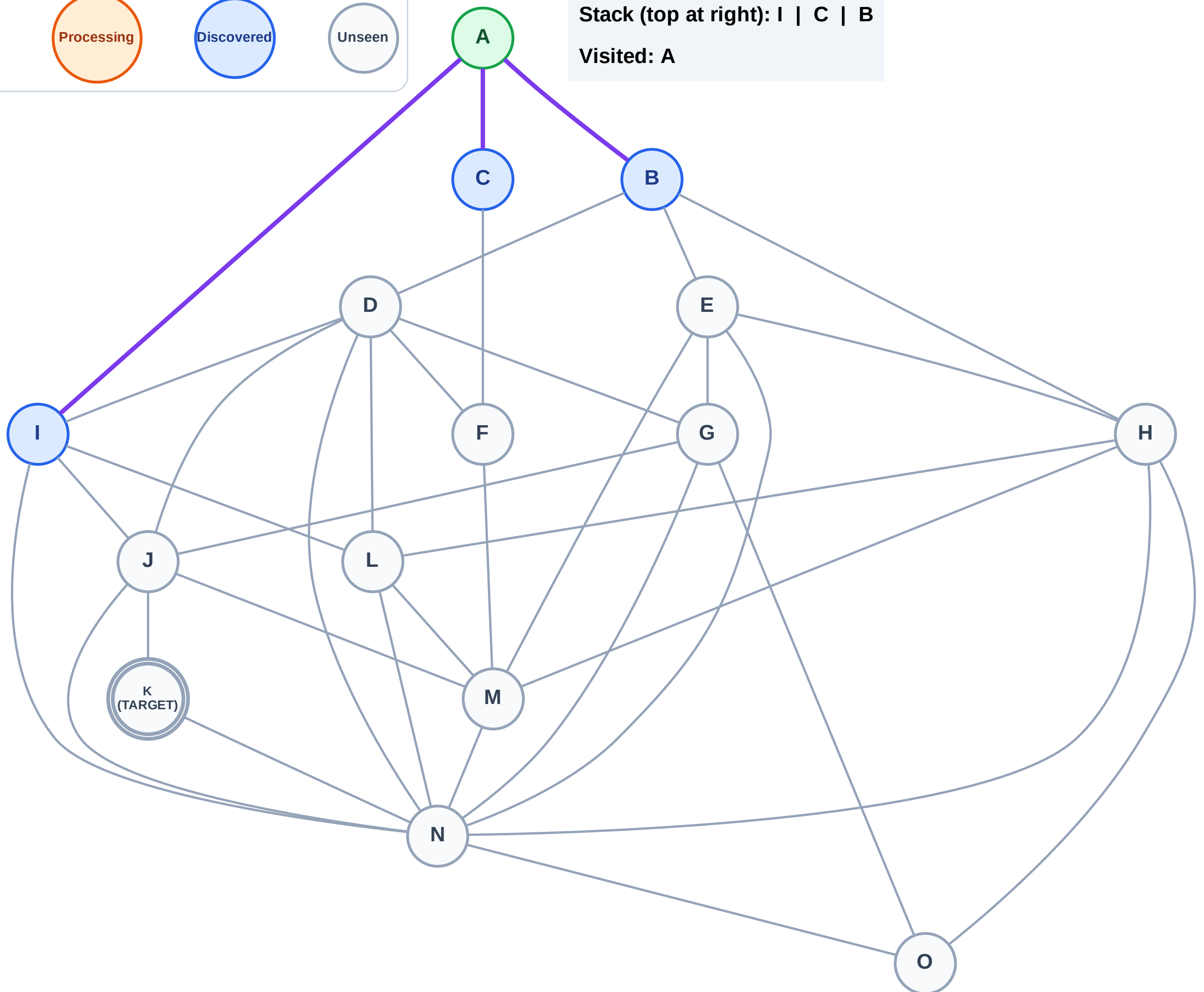
Step 3: Discover I, C, B from A

Legend



Stack (top at right): I | C | B

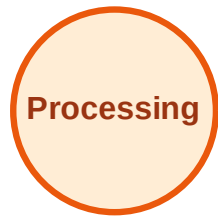
Visited: A



DFS Traversal

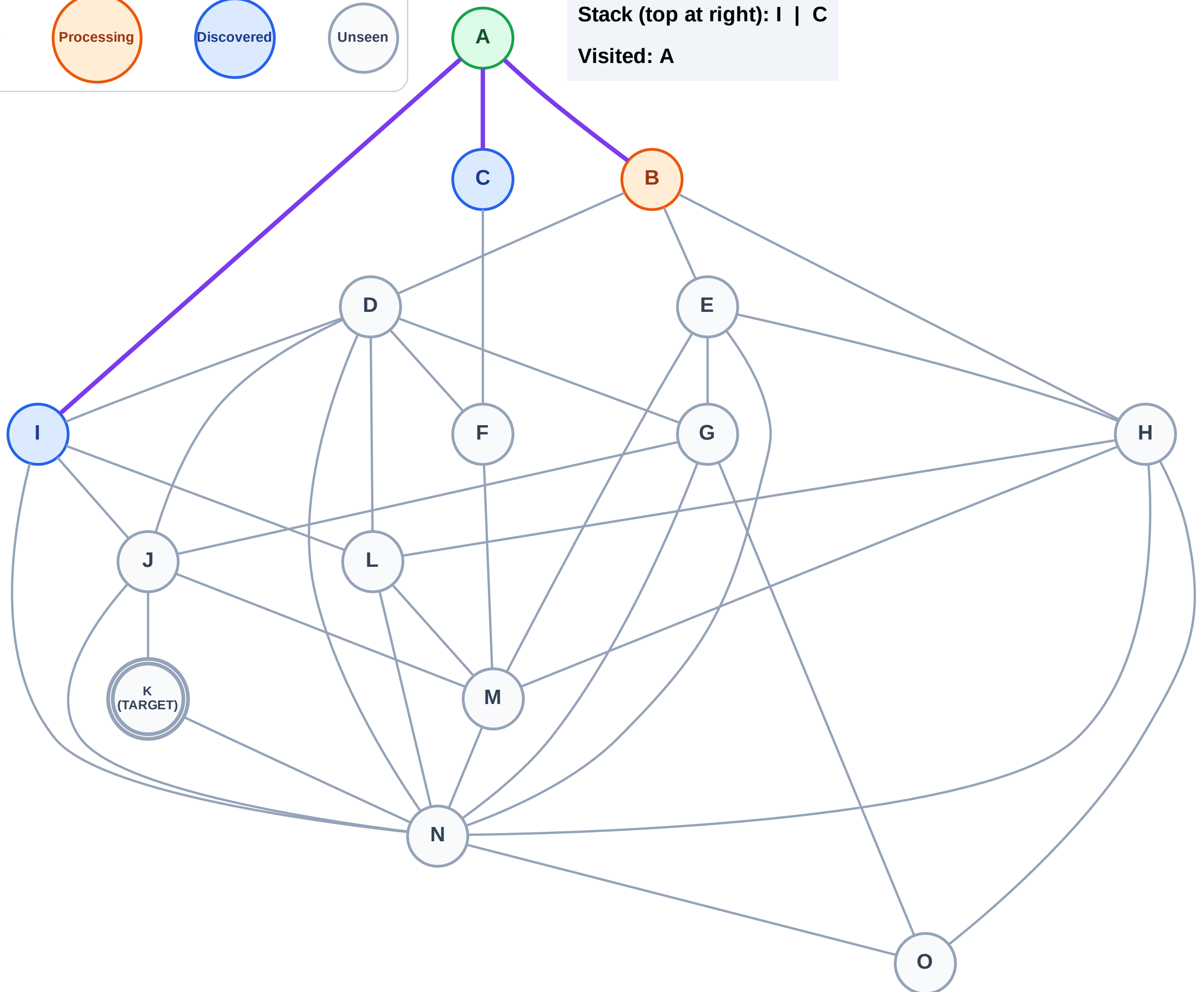
Step 4: Process node B

Legend



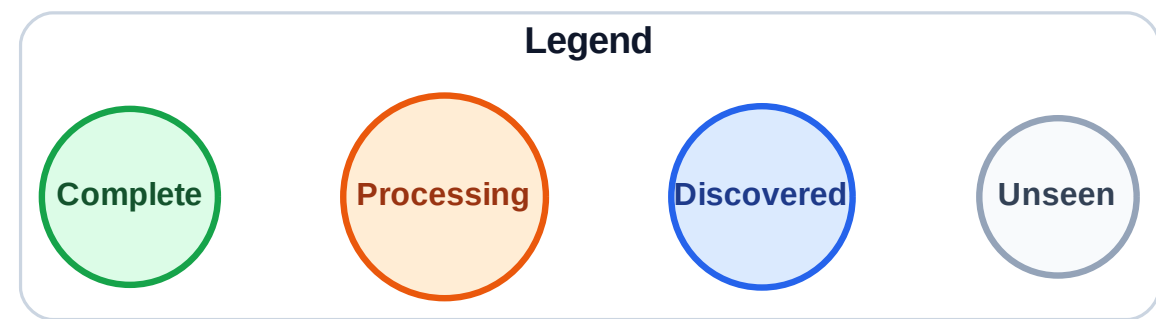
Stack (top at right): I | C

Visited: A



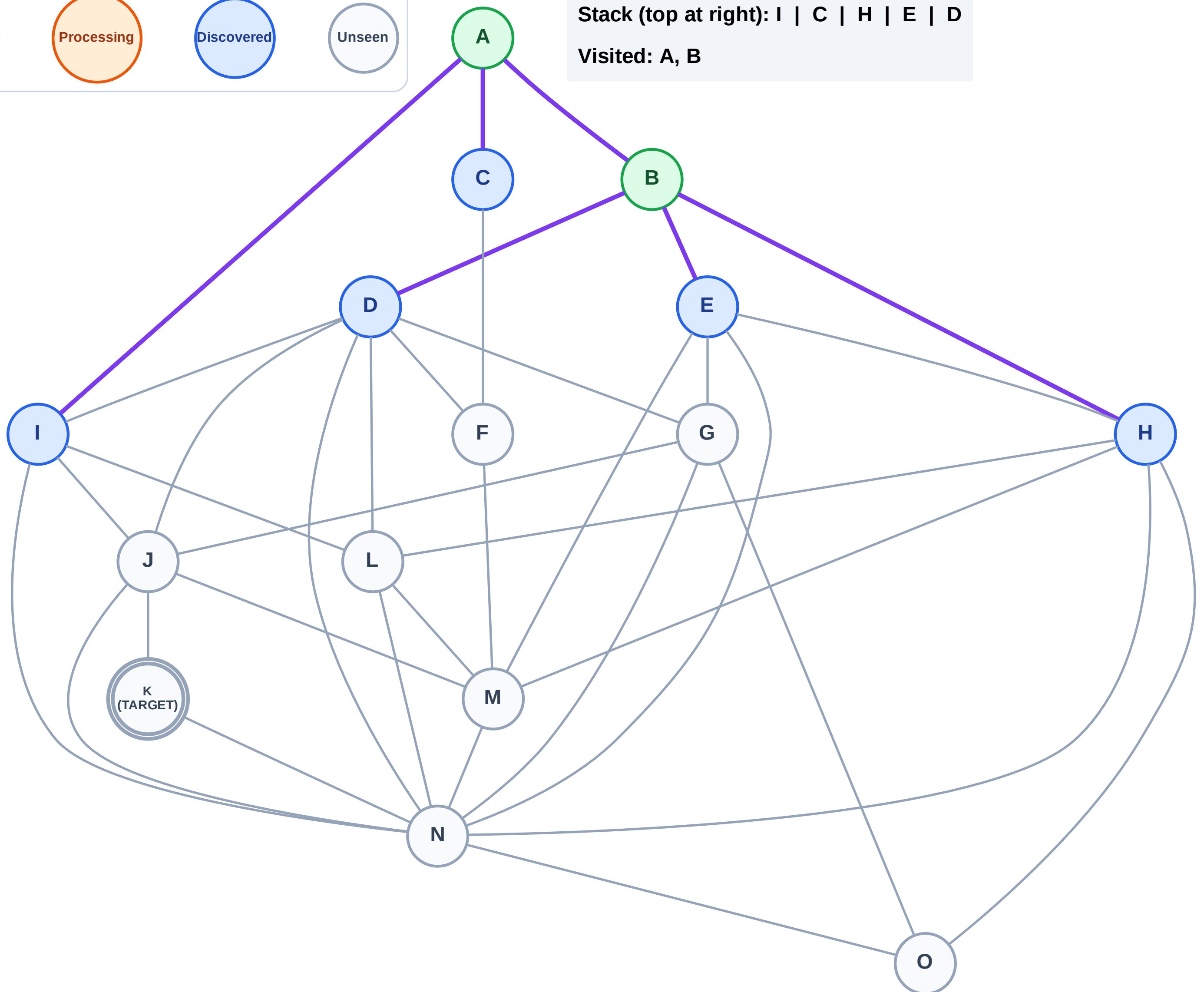
DFS Traversal

Step 5: Discover H, E, D from B



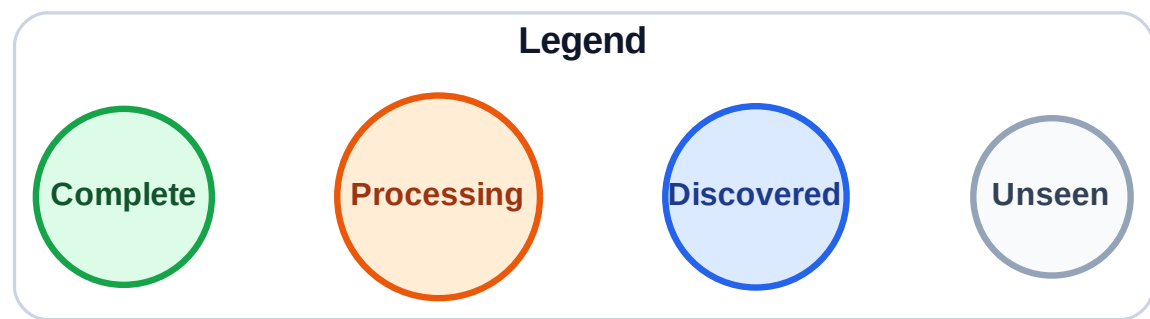
Stack (top at right): I | C | H | E | D

Visited: A, B



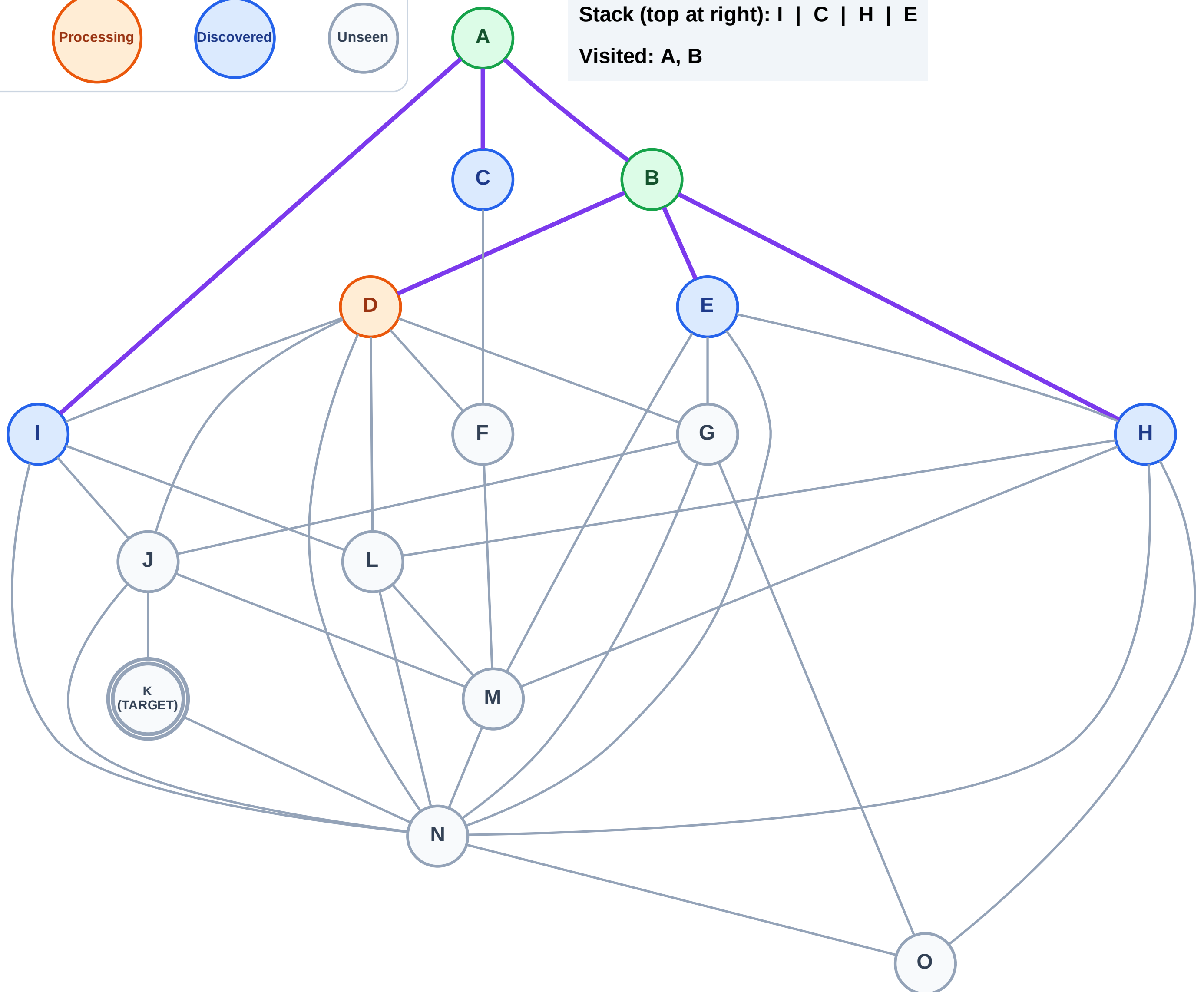
DFS Traversal

Step 6: Process node D



Stack (top at right): I | C | H | E

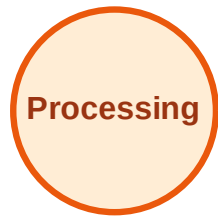
Visited: A, B



DFS Traversal

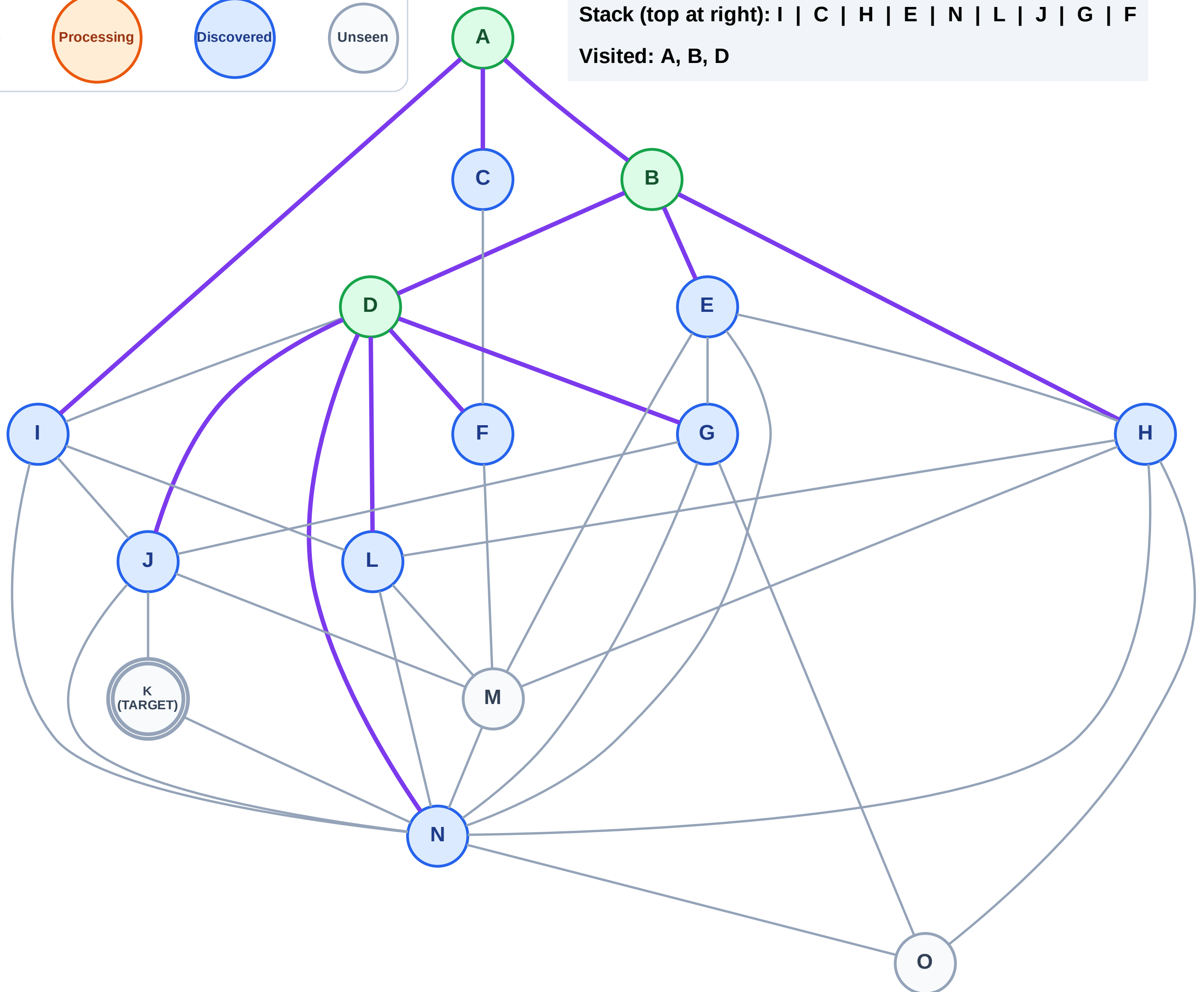
Step 7: Discover N, L, J, G, F from D

Legend



Stack (top at right): I | C | H | E | N | L | J | G | F

Visited: A, B, D



DFS Traversal

Step 8: Process node F

Legend

Complete

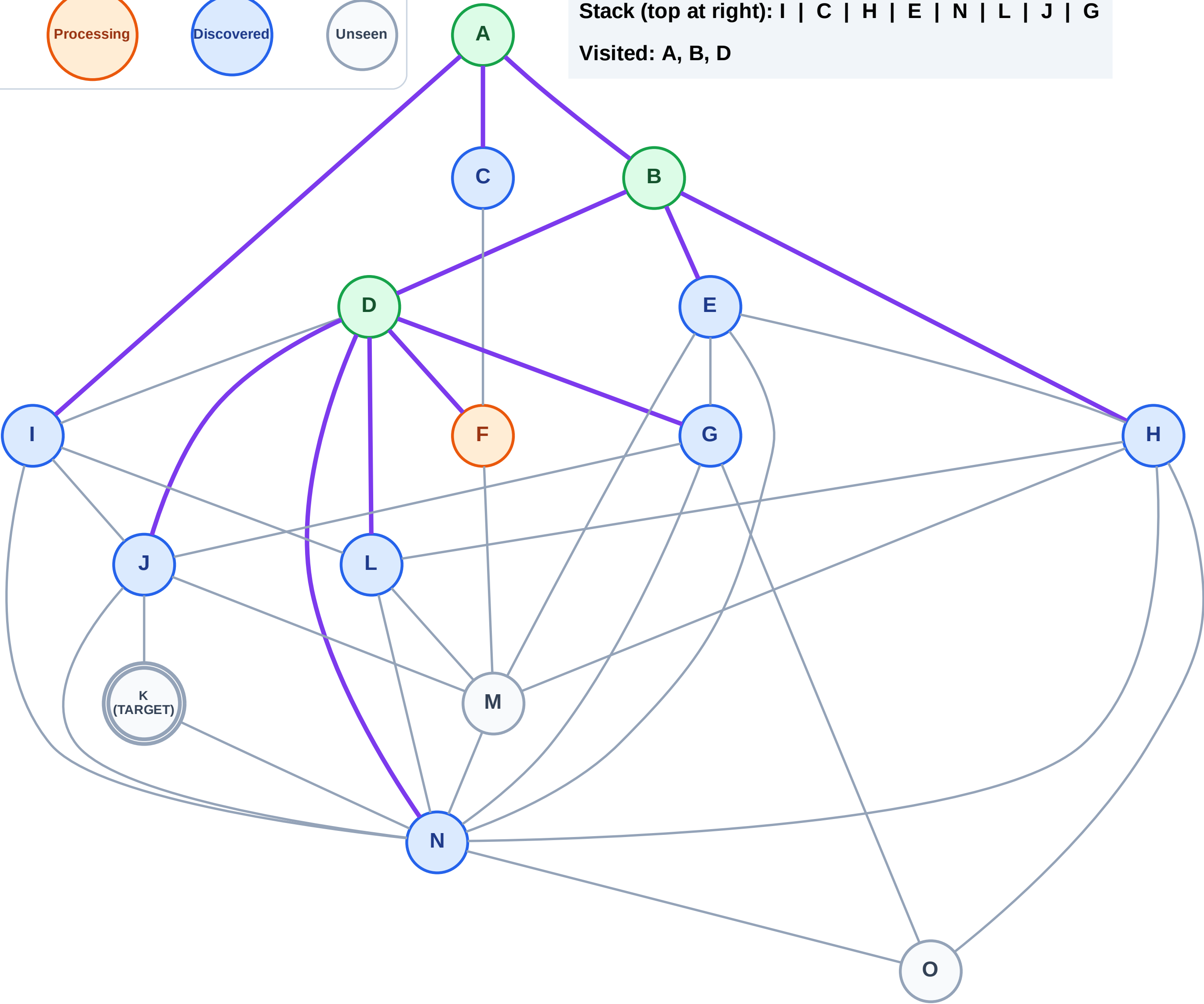
Processing

Discovered

Unseen

Stack (top at right): I | C | H | E | N | L | J | G

Visited: A, B, D



DFS Traversal

Step 9: Discover M from F

Legend



Complete



Processing



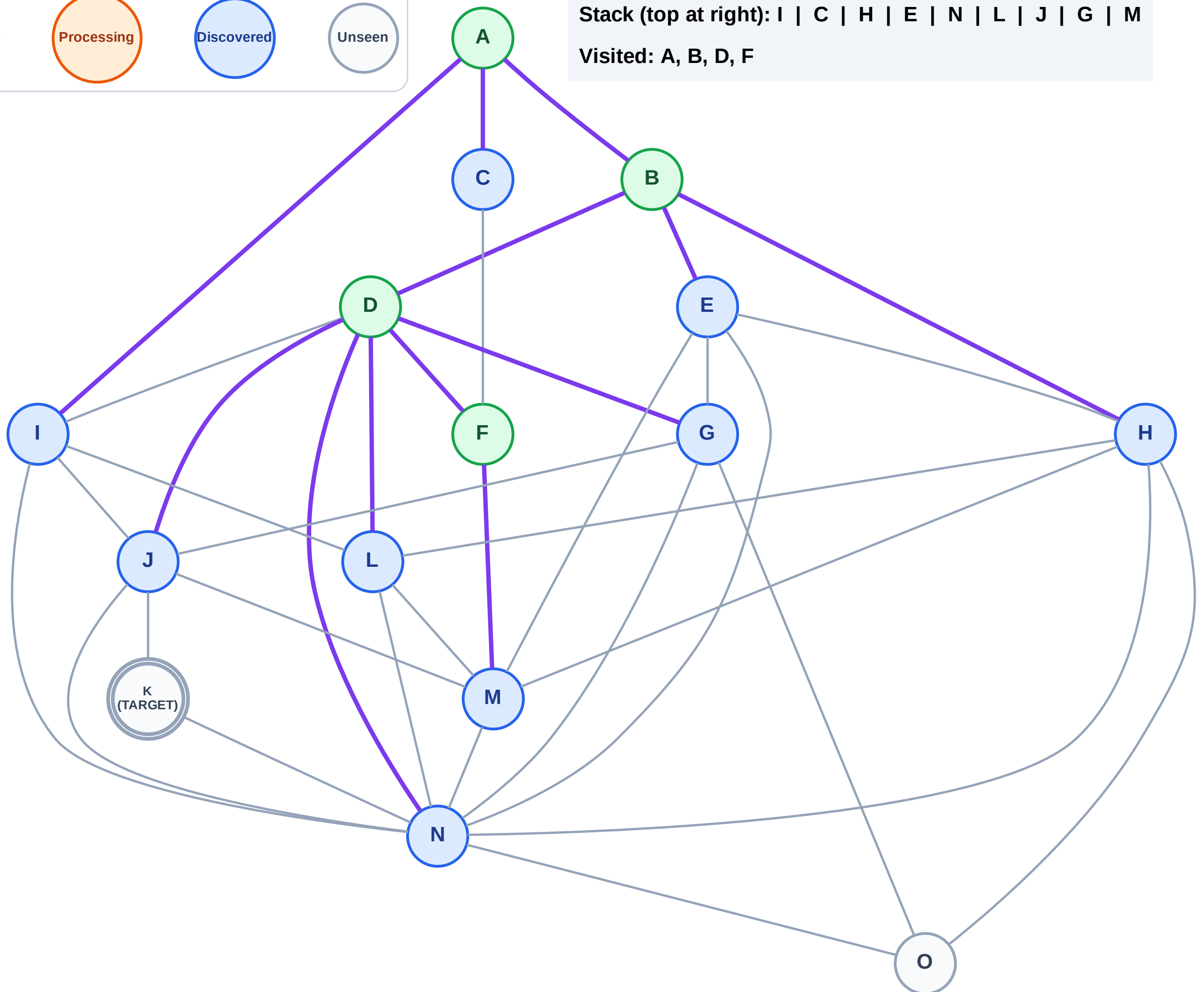
Discovered



Unseen

Stack (top at right): I | C | H | E | N | L | J | G | M

Visited: A, B, D, F



DFS Traversal

Step 10: Process node M

Legend

Complete

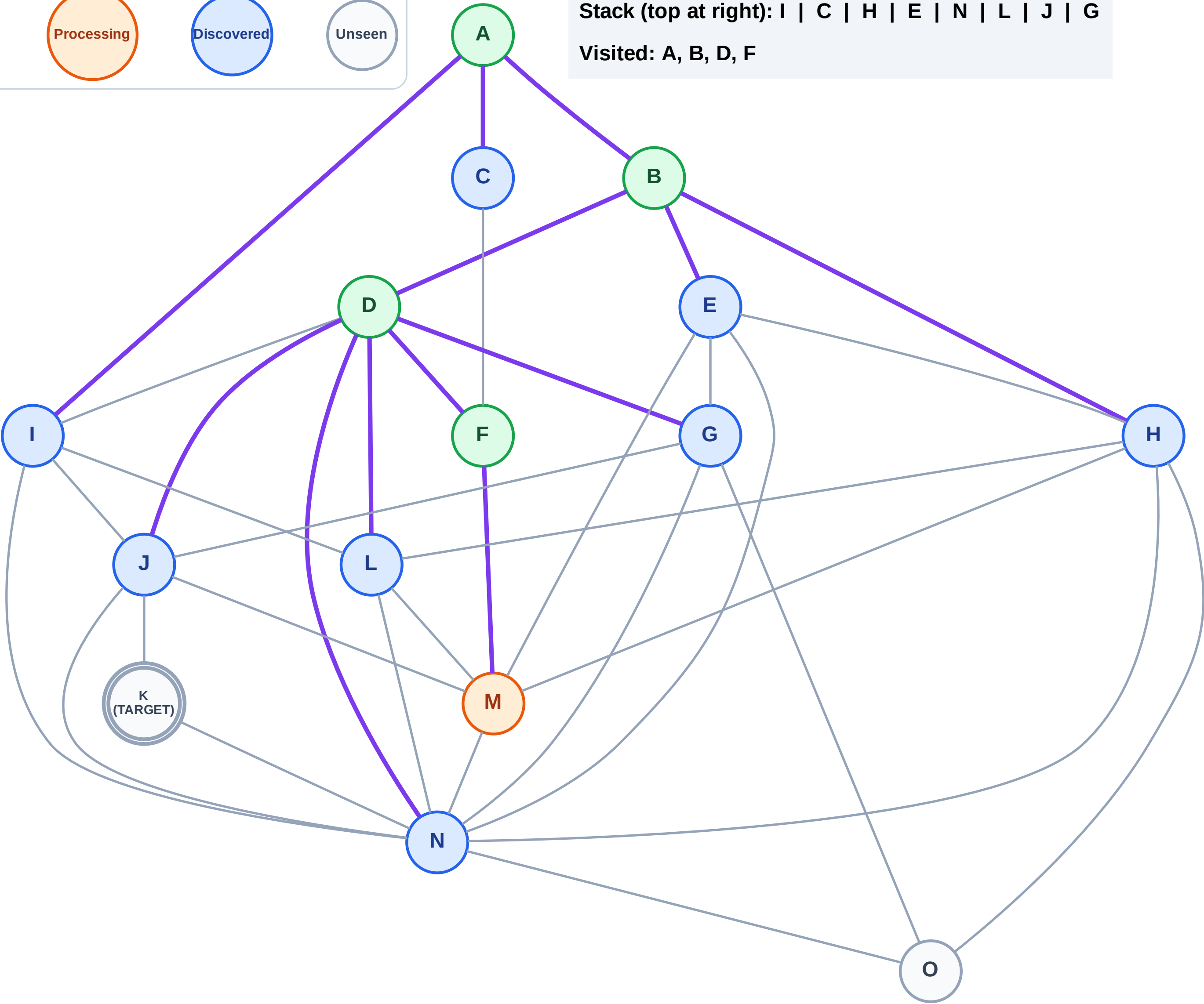
Processing

Discovered

Unseen

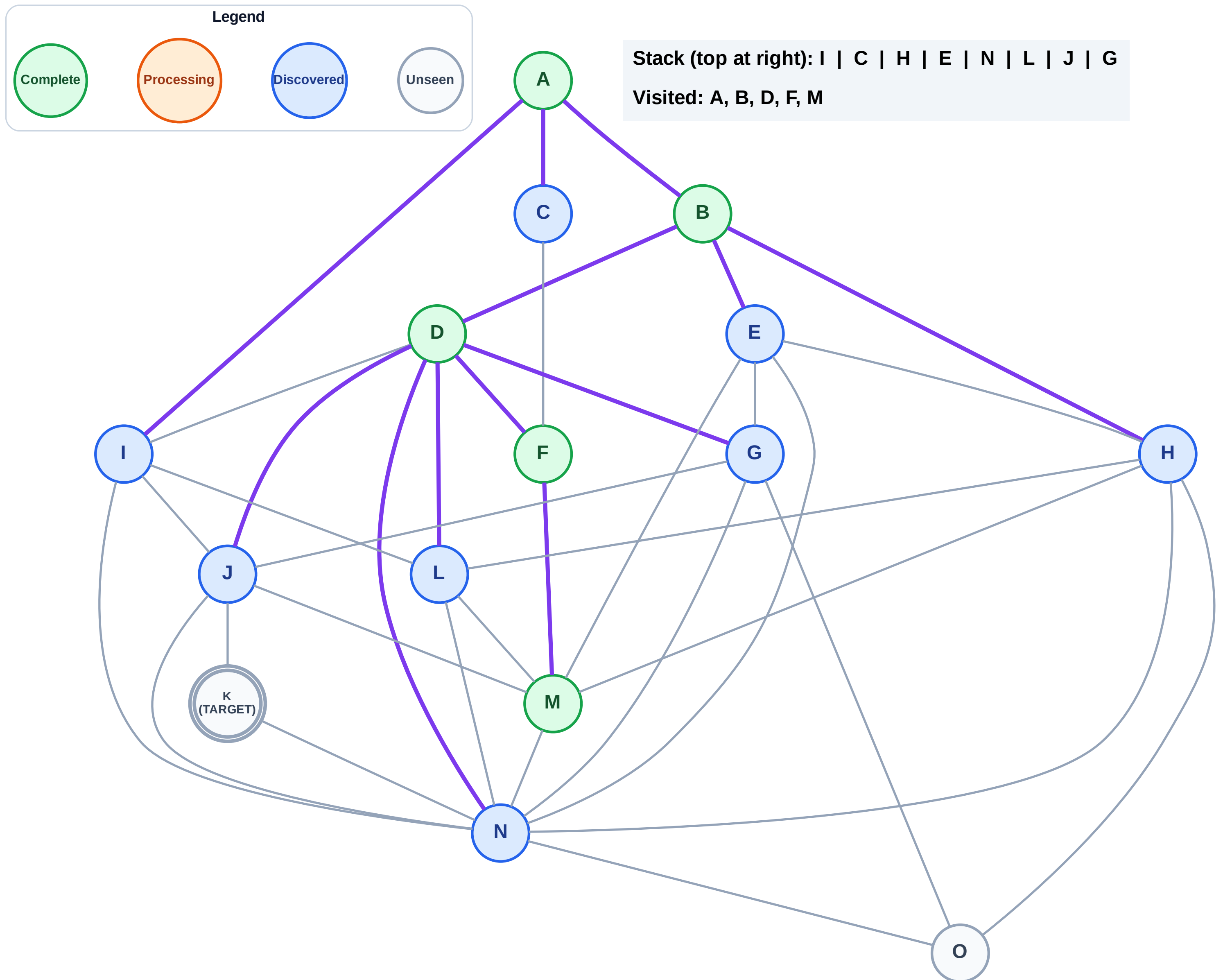
Stack (top at right): I | C | H | E | N | L | J | G

Visited: A, B, D, F



Step 11: No new neighbors from M

Step 11: No new neighbors from M



DFS Traversal

Step 12: Process node G

Legend

Complete

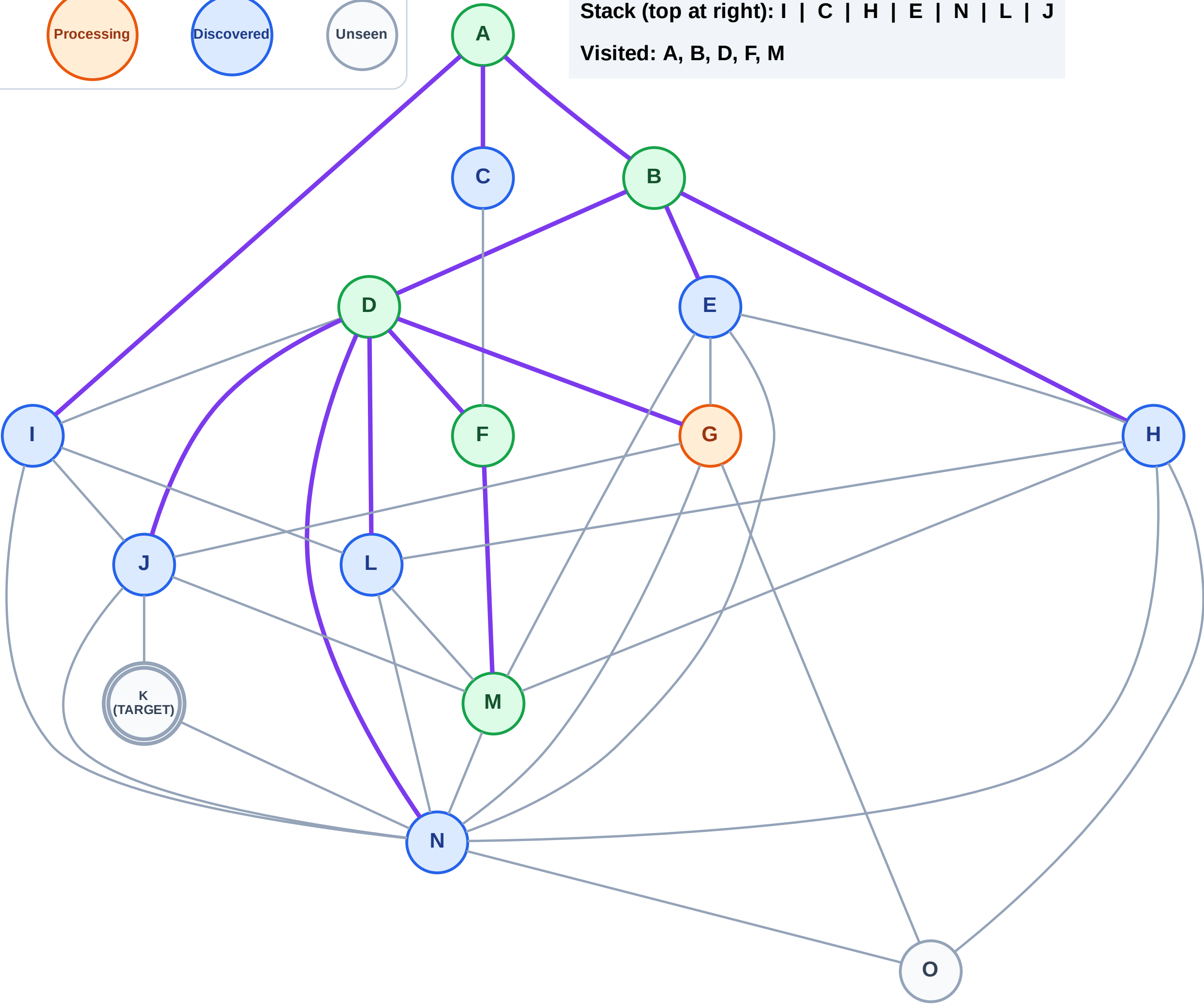
Processing

Discovered

Unseen

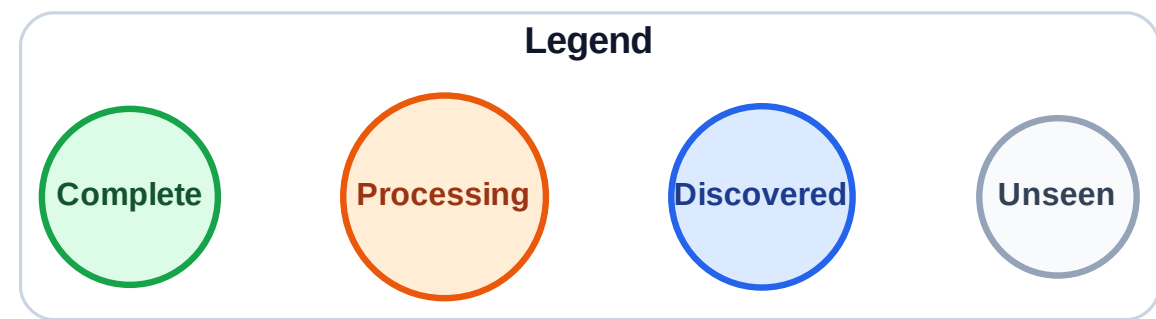
Stack (top at right): I | C | H | E | N | L | J

Visited: A, B, D, F, M



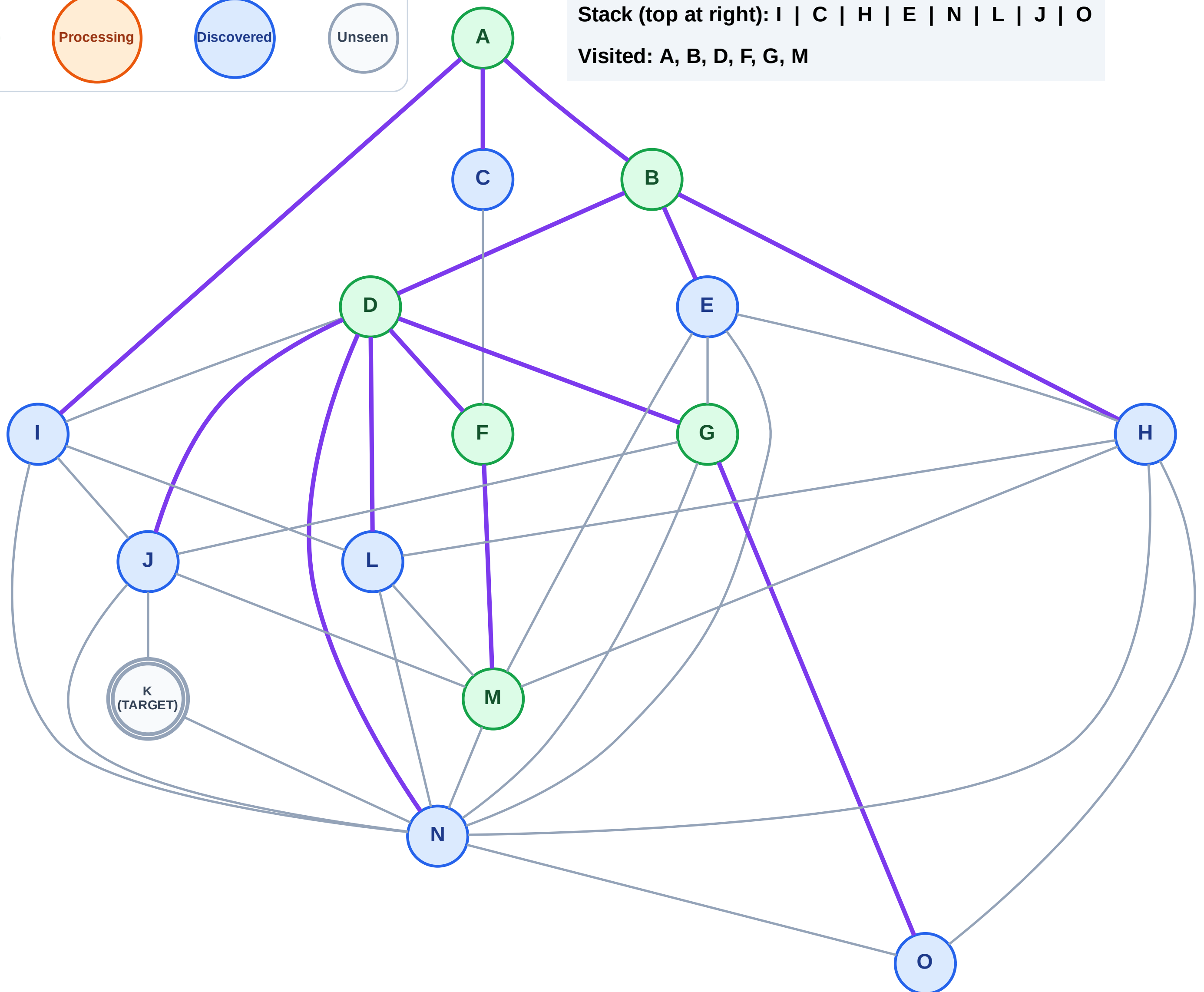
DFS Traversal

Step 13: Discover O from G



Stack (top at right): I | C | H | E | N | L | J | O

Visited: A, B, D, F, G, M



DFS Traversal

Step 14: Process node O

Legend

Complete

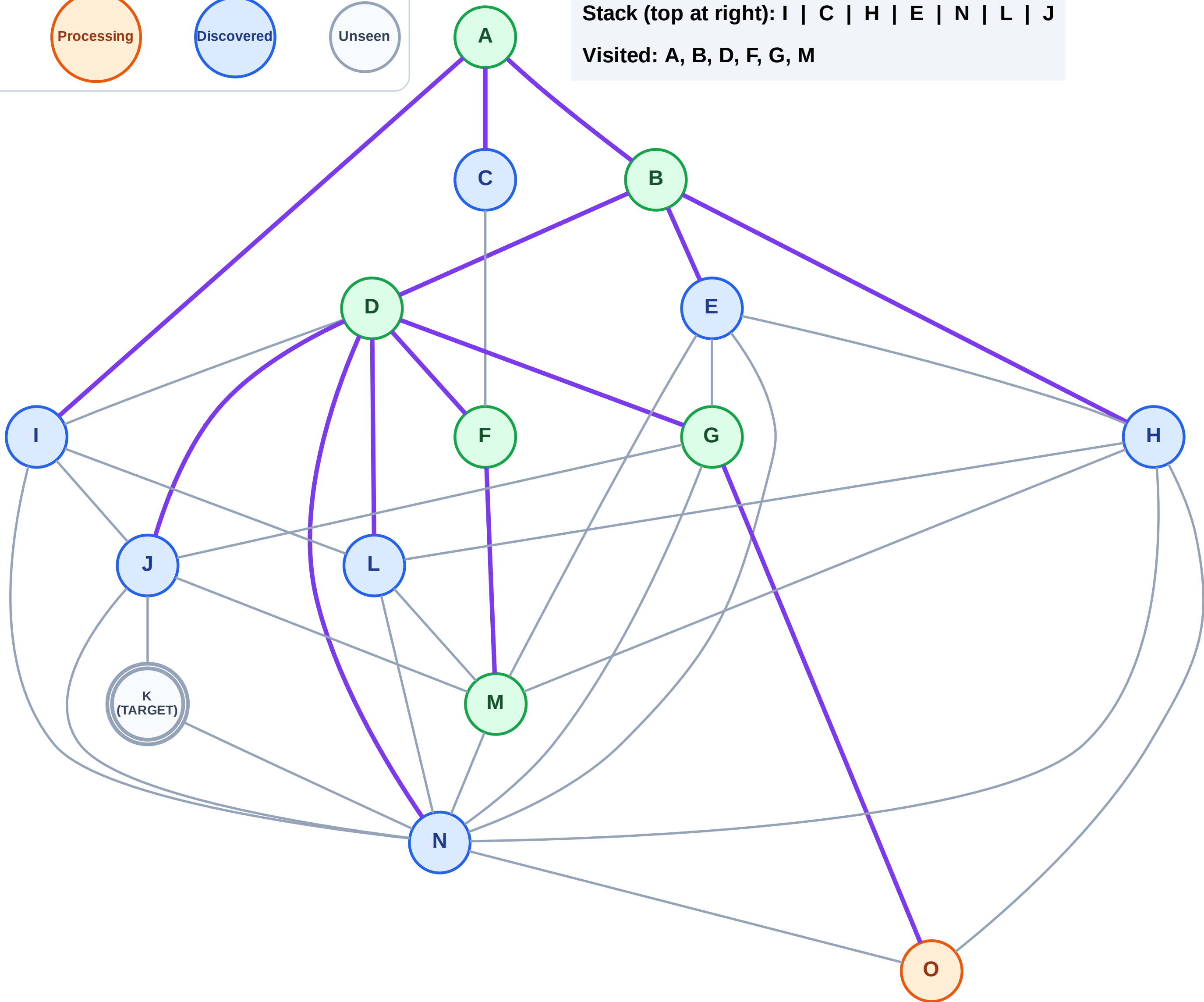
Processing

Discovered

Unseen

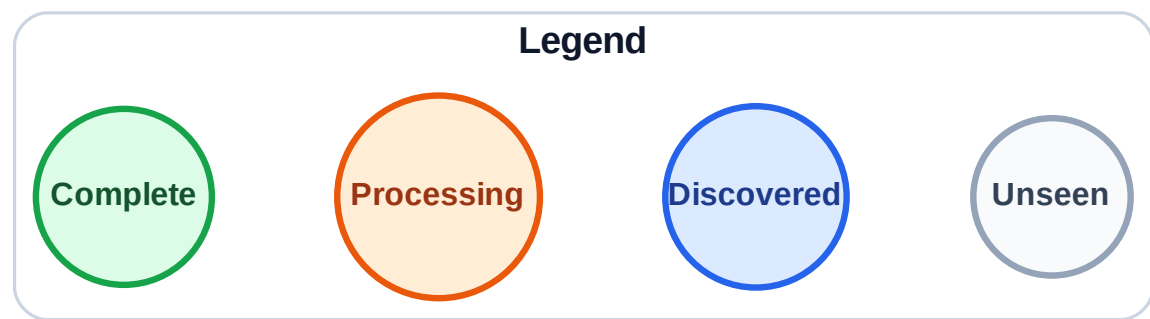
Stack (top at right): I | C | H | E | N | L | J

Visited: A, B, D, F, G, M



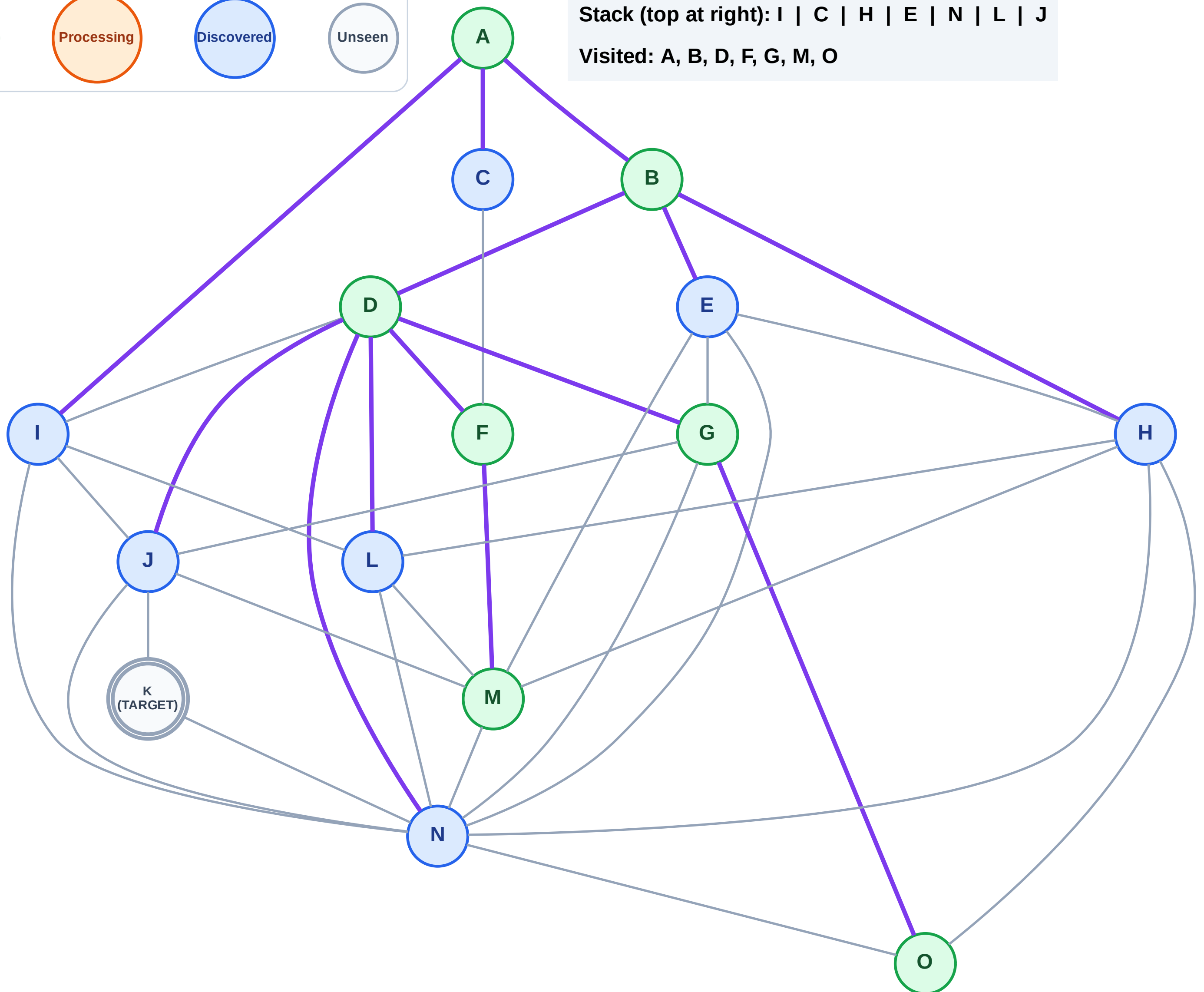
DFS Traversal

Step 15: No new neighbors from O



Stack (top at right): I | C | H | E | N | L | J

Visited: A, B, D, F, G, M, O



DFS Traversal

Step 16: Process node J

Legend



Complete



Processing



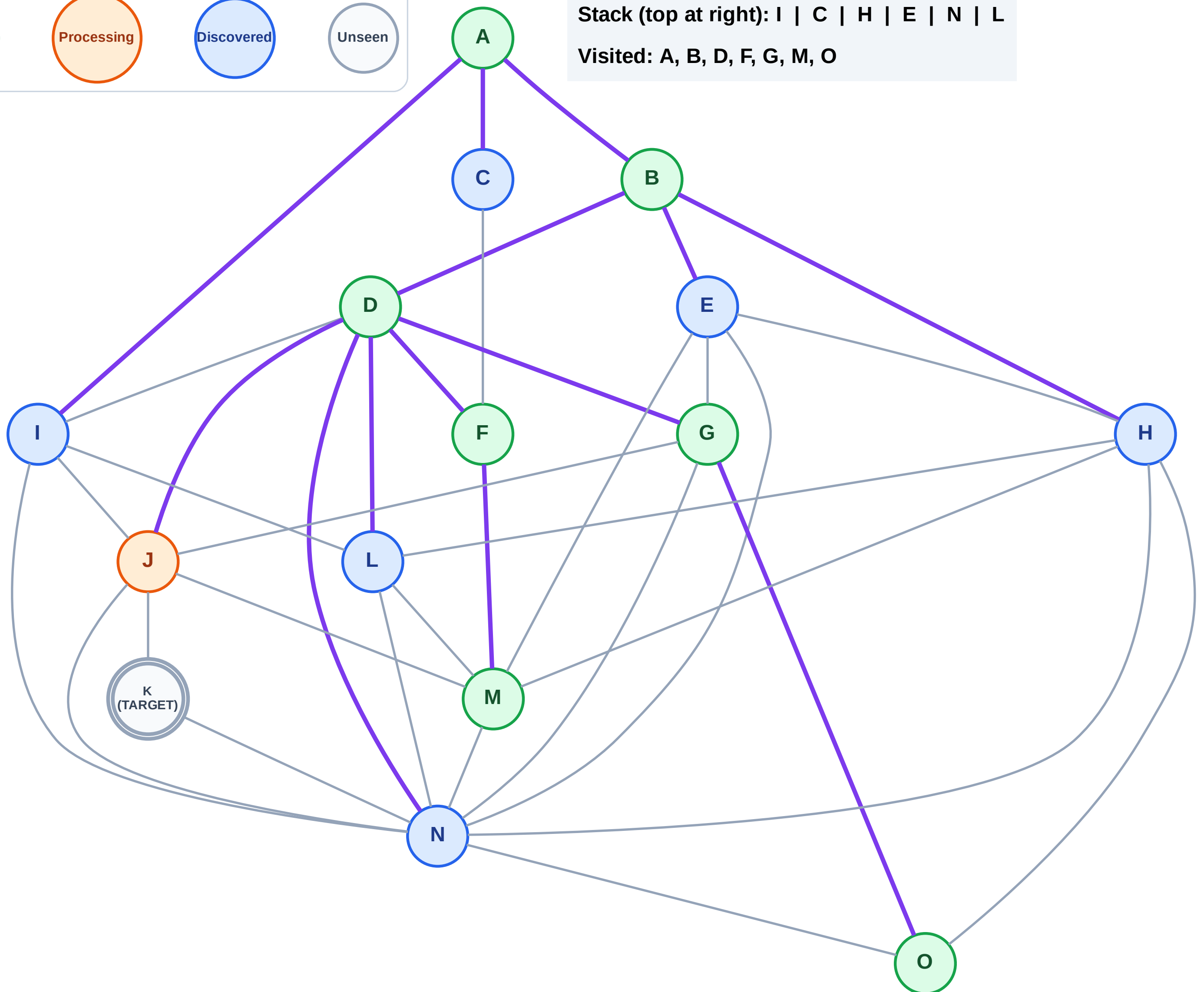
Discovered



Unseen

Stack (top at right): I | C | H | E | N | L

Visited: A, B, D, F, G, M, O



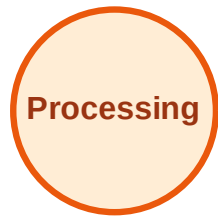
DFS Traversal

Step 17: Discover K from J

Legend



Complete



Processing



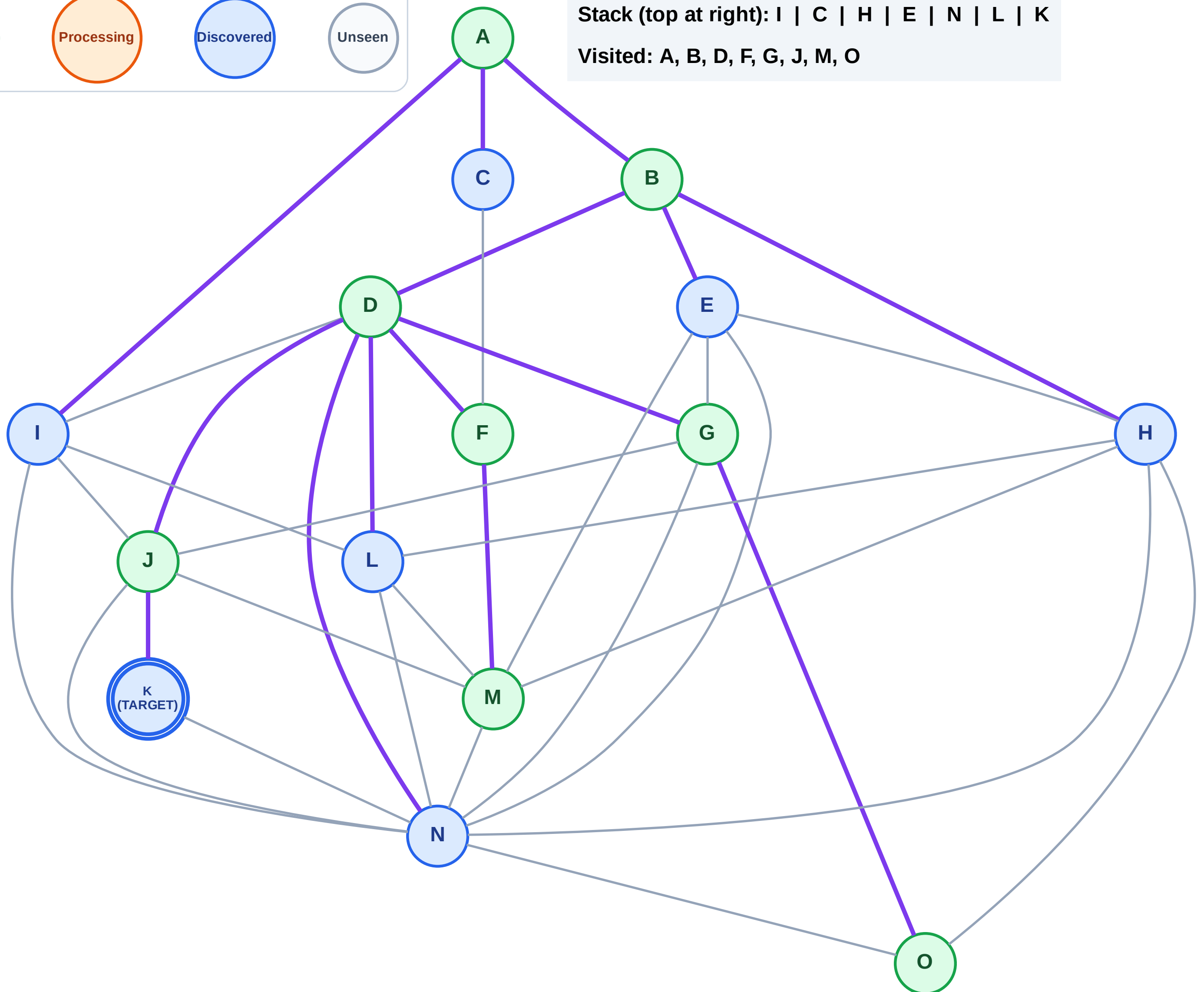
Discovered



Unseen

Stack (top at right): I | C | H | E | N | L | K

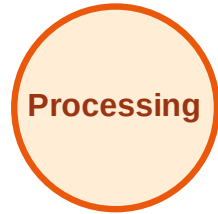
Visited: A, B, D, F, G, J, M, O



DFS Traversal

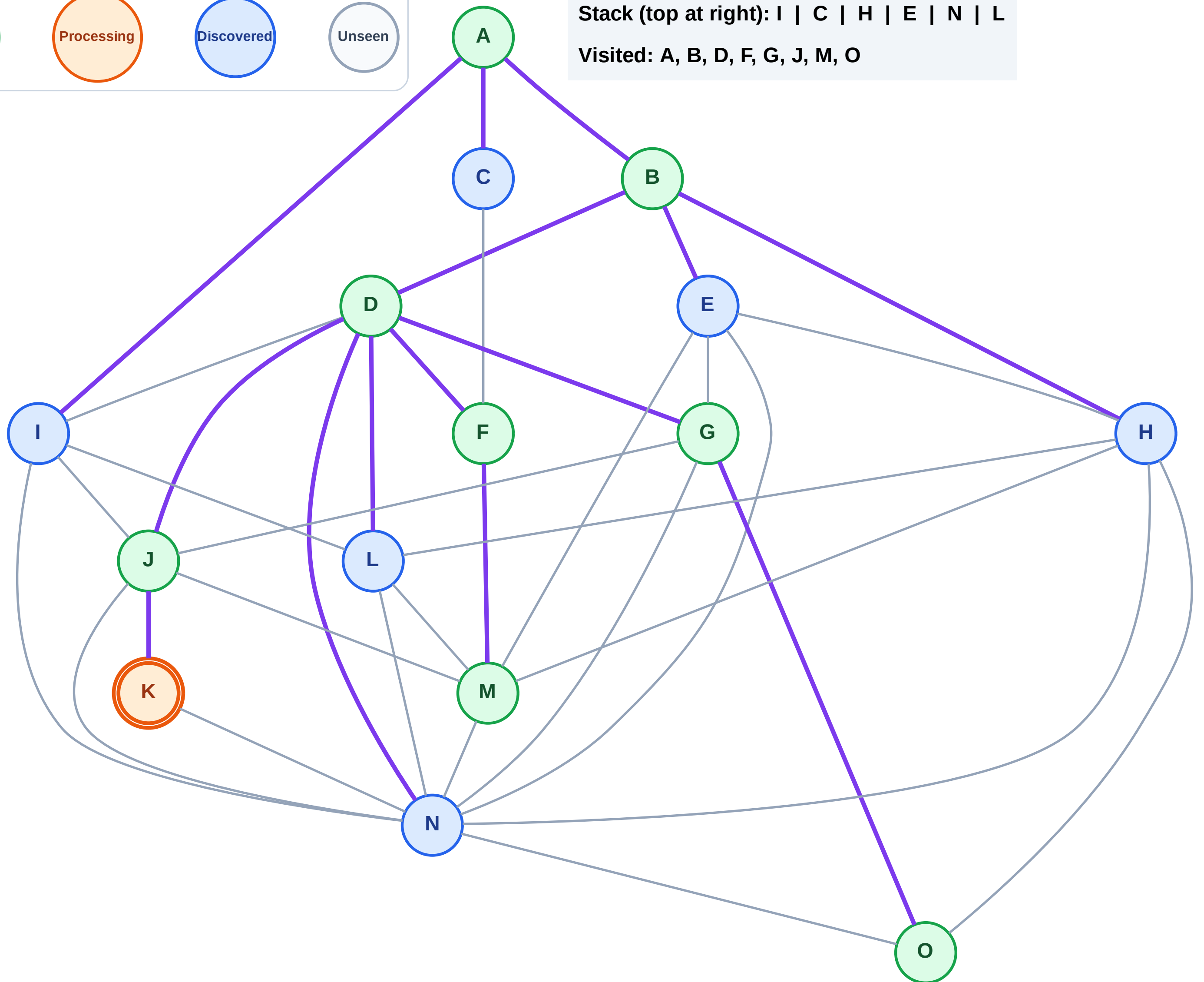
Step 18: Process node K

Legend



Stack (top at right): I | C | H | E | N | L

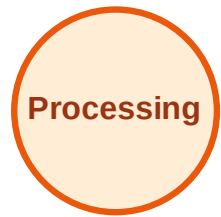
Visited: A, B, D, F, G, J, M, O



DFS Traversal

Step 19: TARGET NODE K REACHED! Path (4 hops): A → B → D → J → K

Legend



Stack (top at right): I | C | H | E | N | L

Visited: A, B, D, F, G, J, K, M, O

